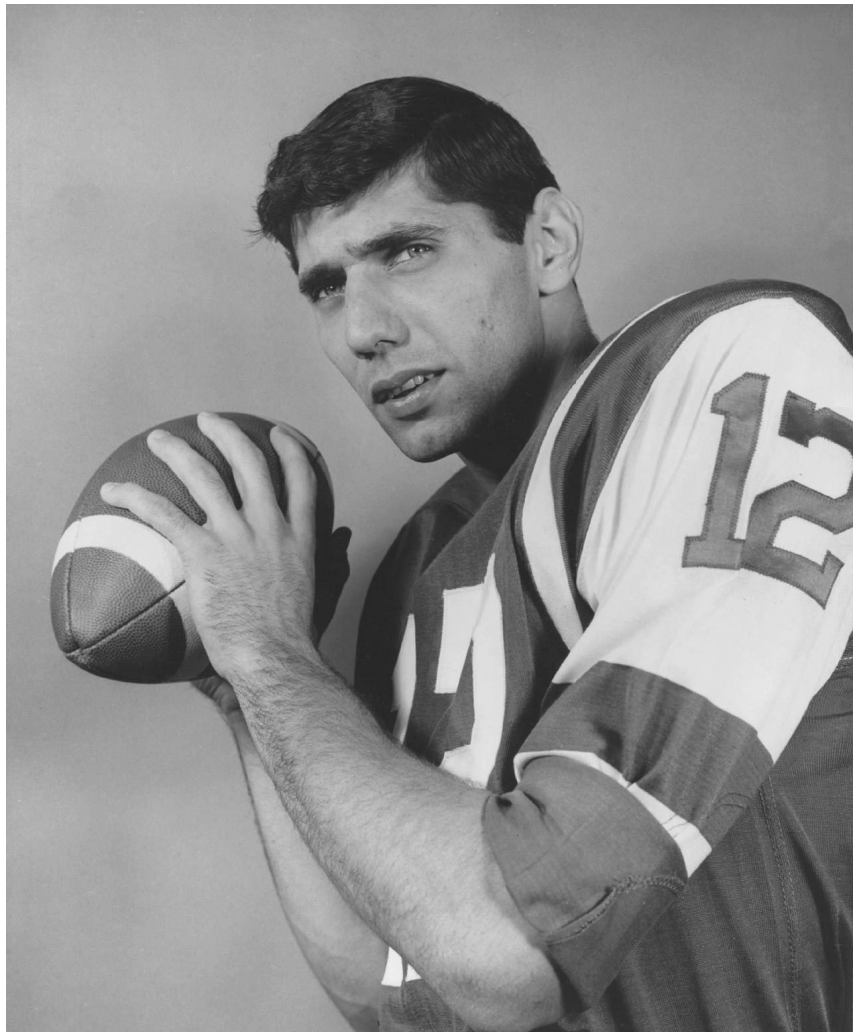


Photo: Joe Namath. New York Jets; Public domain.

BEHAVIORAL NEUROSCIENCE 392

Special Topic: Cognitive and Clinical Neuroscience

Football, Memory, and Mental Health: Players, Brain Imaging, and CTE

Fall 2026

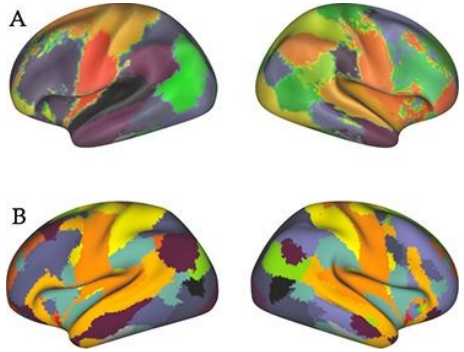
Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering

Football requires perception, memory, and control

A football player must identify movement, remember the play, and choose an action while other people are moving. These demands involve several cognitive systems. Cognition means the processes used to acquire information, maintain it, and use it to guide behavior, including attention, memory, language, and decision-making.

The Michigan and Northwestern recording shows players blocking, pursuing, and changing direction during real plays. Their visible actions provide examples of these demands. A completed pass or missed assignment is a sporting outcome; a clinical assessment uses controlled tasks to examine particular abilities more directly.

After a playing career, evaluation asks whether a person's abilities have changed from their earlier level. It also asks how any difficulty affects work, relationships, and independence. Chronic traumatic encephalopathy, or CTE, names a particular brain disease; clinical symptoms require their own description and assessment.



Colclough et al. (2017), Fig. 2.



Michigan–Northwestern football plays. FOX Sports (2018).

YouTube source

Play video

Mike Webster's case began with changes in functioning

Mike Webster played center, an offensive position that requires remembering assignments and coordinating closely with teammates. Public accounts and his original case report described later cognitive and behavioral difficulties. A clinical history records what changed, when it changed, and how the changes affected ordinary activities outside football.

Memory complaints can describe different failures. A person may fail to attend when information is presented, forget information after learning it, or struggle to retrieve an available memory. These processes can produce similar everyday complaints but require different questions and tasks during a cognitive evaluation.

Webster's postmortem examination identified abnormal tau pathology in brain tissue. The tissue finding and the account of his functioning were different observations, collected through different methods. Modern studies assess larger groups to examine which clinical features predict pathology and which also occur with other conditions.

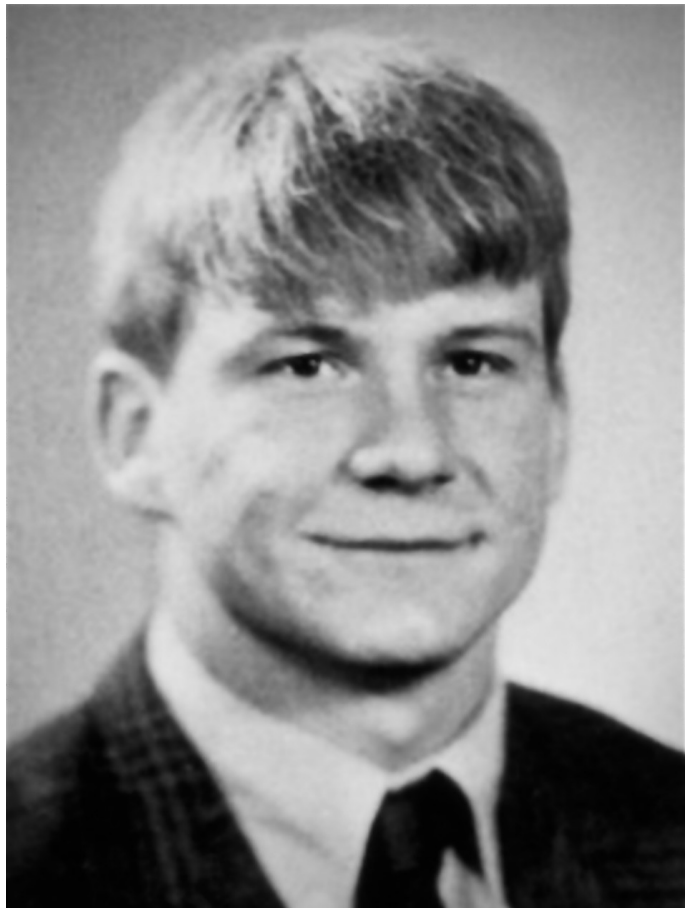


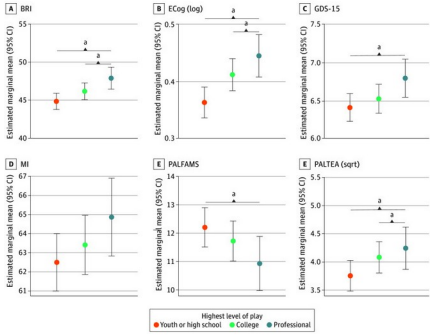
Photo: Mike Webster. Rhineland High School.; Public domain.

Former players showed differences on cognitive testing

A 2026 study examined cognitive and neuropsychiatric function in former American football players. Participants completed computerized testing and reported symptoms. Higher levels of play and more years of football were associated with worse cognitive performance and greater neuropsychiatric symptoms within this observational sample.

The investigators also compared a subset of former players with matched controls. Players performed worse on a computerized cognitive measure and reported more cognitive concerns and depressive symptoms. A performance measure records behavior under specified testing conditions, whereas a symptom questionnaire records the person's experience of difficulty.

The examination recording demonstrates orientation and memory questions in a separate clinical teaching setting. The examiner asks about time, place, and remembered information. These are examples of observable responses used in assessment; the person in the recording is not identified as a football player or a participant in the cohort.



Aaronson et al. (2026), Fig. 2.



Clinical demonstration: orientation and memory. Larsen and Stensaas (2020).

Play video

Junior Seau's history and tissue were different evidence

Junior Seau played linebacker, a role involving pursuit, rapid responses, and physical contact. An examination of his donated brain by National Institutes of Health researchers identified CTE after death. His public case increased attention to the experiences of former players and the need to investigate clinical changes carefully.

A mood symptom describes an emotional experience such as persistent sadness or loss of interest. A behavioral symptom describes an action or difficulty regulating actions. A cognitive symptom describes a problem with processes such as remembering or planning. These categories can overlap, but they are not interchangeable descriptions.

A confirmed tissue diagnosis does not assign a single cause to every event in a person's life. Clinical evaluation examines symptom timing, other illnesses, sleep, pain, and medication exposure. The relevant psychological questions concern specific abilities and experiences rather than conclusions drawn from the player's photograph or career.



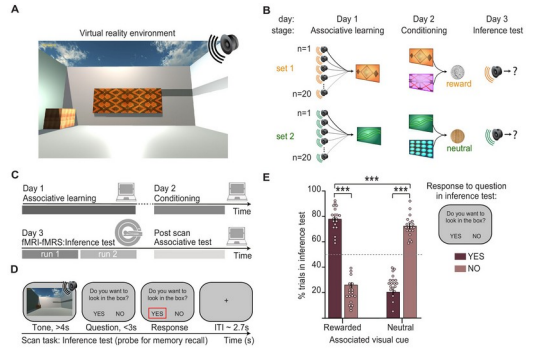
Photo: Junior Seau. Dave Sizer; CC BY 2.0.

Working memory holds information while it is used

Working memory maintains information briefly while the person performs an operation on it. Remembering a phone number long enough to enter it is one example. Repeating numbers backward adds another demand because the person must preserve the sequence and reorganize its order before giving an answer.

The clinical demonstration includes backward digit span, backward spelling, and reciting months in reverse. These tasks require attention to the instruction and continued maintenance of the goal. Losing the instruction, dropping an item, and producing the wrong order are different errors that the examiner can record.

Frontal and parietal systems participate in maintaining goals and manipulating information. Working memory therefore differs from recalling an event that happened yesterday. A player may have difficulty in one process while performing adequately in another, so an assessment samples several abilities instead of treating memory as a single score.



Koolschijn et al. (2021), Fig. 1.



Clinical demonstration: working memory. Larsen and Stensaas (2020).

Play video

Ken Stabler's family described everyday memory changes

Ken Stabler played quarterback and later worked in sports broadcasting. The Boston University CTE Center describes his family's reports of repetition, confusion while driving, and changes they noticed over time. After his death, examination of his donated brain identified CTE. The report separates lived observations from tissue analysis.

Driving requires remembering a destination, recognizing the surroundings, following rules, and changing plans when traffic changes. Confusion during driving can therefore arise from more than one cognitive difficulty. An examiner asks for concrete examples and determines whether the problem is new rather than inferring the impaired process from one description.

Family observations can add information about changes that are difficult to capture in a brief office visit. They also help establish a timeline of independence and assistance. Clinical testing then measures specific abilities, allowing the history and the test results to describe complementary parts of the person's functioning.



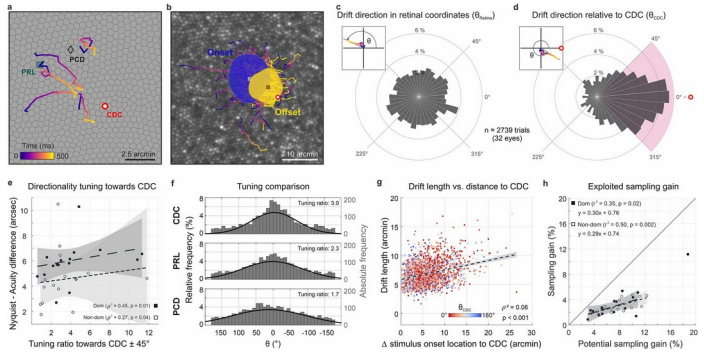
Photo: Ken Stabler. NOAA; Public domain.

Tiny eye movements improve visual sampling

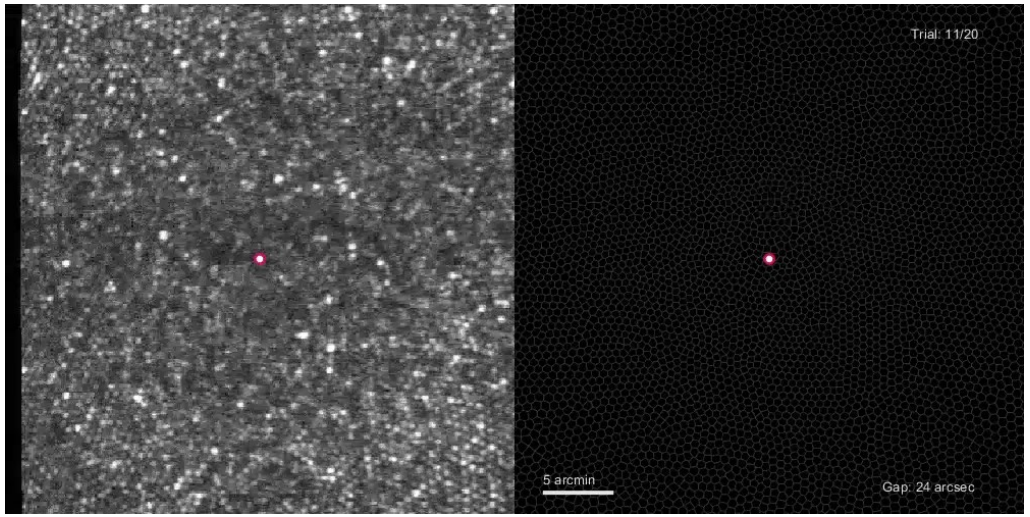
Researchers imaged individual cone photoreceptors in the human retina while measuring visual resolution. Cones are light-sensitive cells that sample the visual scene. Even when participants attempted to fixate, their eyes continued to drift, moving the stimulus across different parts of the receptor array.

The original recording shows the stimulus moving across the retina alongside a representation of cone activation. The study found that the direction and length of this drift contributed to resolution. The eyes adjusted sampling toward regions with denser cones within a short interval during viewing.

Recognizing a teammate or reading a test item begins with sensory acquisition before memory and decision processes can operate. This experiment measured visual sampling in human volunteers, not CTE. Poor performance on a visually presented cognitive task can require examination of vision as well as the cognitive ability being tested.



Witten et al. (2024), Fig. 4.



Retinal sampling during fixation. Witten et al. (2024).

Play video

Frank Gifford's family reported a postmortem finding

Frank Gifford played for the New York Giants before a long broadcasting career. In 2015, his family publicly reported that examination after his death found CTE. Their statement did not provide a detailed clinical symptom inventory, so the public record should not be expanded into an invented psychological profile.

A clinical profile describes which abilities are impaired and which remain relatively preserved. It can include memory, attention, language, regulation of behavior, and the ability to manage daily tasks. Two people with the same pathological diagnosis can differ in the severity and distribution of their difficulties.

Descriptions from families, medical records, and formal tests provide different kinds of information. A study can compare these observations with brain tissue when both are available. Such comparisons help identify useful clinical features without assuming that every person with a particular diagnosis followed an identical sequence of symptoms.



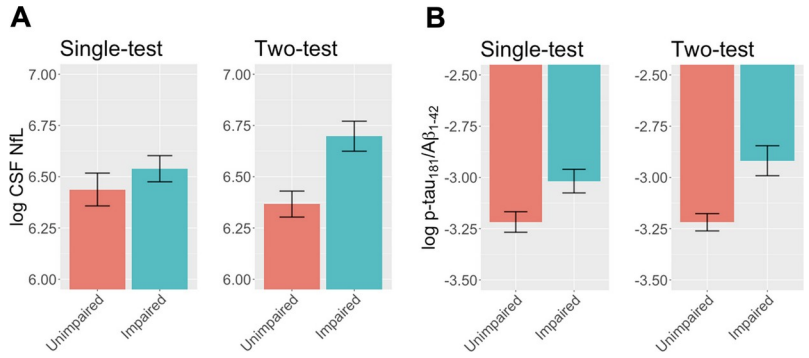
Photo: Frank Gifford. Howardform at English Wikipedia; Public domain.

A low score changes meaning across test definitions

Investigators compared two ways of defining cognitive impairment in former football players. One approach used a sufficiently low result on a single test. The other required lower performance on two tests within the same domain, such as memory or executive functioning, before classifying the domain as impaired.

The two-test definition identified fewer participants and showed associations with cerebrospinal-fluid markers of neurodegeneration that the single-test definition did not. An association with neurofilament light remained after excluding amyloid-positive participants. The paper examined how an assessment rule changes the biological characteristics of the group it identifies.

The separate examination recording shows a person who makes errors on orientation and memory questions. It demonstrates the behavior an examiner records, rather than the study's classification procedure. A visible error is an observation; deciding whether a cognitive domain is impaired requires the appropriate tests, norms, and clinical context.



Ly MT et al. (2025), Fig. 1.



Clinical demonstration: orientation errors. Larsen and Stensaas (2020).

Play video

Nick Buoniconti described difficulties before brain donation

Nick Buoniconti played linebacker and became a prominent advocate for neurological research. In a 2017 public report, he described memory loss and difficulties with everyday functioning and pledged to donate his brain. That report described his experiences during life rather than a confirmed postmortem diagnosis at that time.

A subjective complaint is a difficulty reported by the person experiencing it. Examples include losing the thread of a conversation or feeling less mentally efficient. The complaint may be important even when its severity differs from a test score because ordinary life places demands that a brief structured task may not reproduce.

An evaluation compares reported experiences with direct testing and observations from other people. It also considers sensory or movement problems that can interfere with daily actions. Buoniconti's account included difficulty using his left hand, showing why functional dependence cannot automatically be attributed to memory alone.



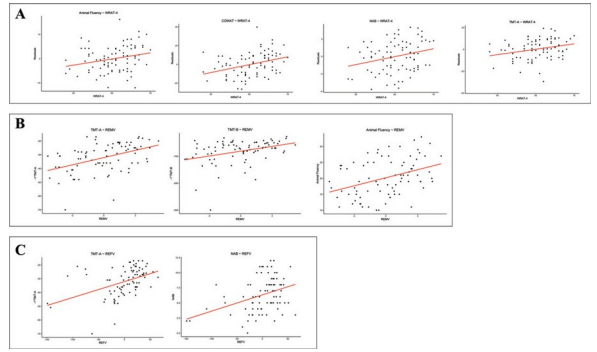
Photo: Nick Buoniconti. Roy Erickson; Public domain.

Reading ability helped explain differences among players

A study of symptomatic former National Football League players examined cognitive reserve. Reserve describes differences in how well people function despite measured brain changes. The researchers tested several proposed indicators, including education, reading ability, and measures derived from performance that remained unexplained by demographic and brain variables.

Higher reading ability was associated with better attention and processing, episodic memory, verbal fluency, and behavioral regulation. Other conventional indicators did not show the same pattern. Reading ability and years of education therefore did not operate as interchangeable measures of reserve in this particular player sample.

The clinical recording demonstrates category-word generation, a verbal-fluency task. The person must retrieve words while maintaining the category and avoiding repetitions. This is a separate teaching demonstration, but it makes one of the study's measured abilities recognizable as an action rather than an abstract label on a results table.



Foley ÉM et al. (2022), Fig. 1.



Clinical demonstration: category-word generation. Larsen and Stensaas (2020).

Play video

Brett Favre reported a different neurological diagnosis

Brett Favre publicly disclosed a Parkinson's disease diagnosis in 2024, as reported by the Boston University CTE Center. Parkinson's disease and CTE are different diagnoses. A history of football participation does not make every later neurological symptom or diagnosis an instance of the same underlying disease.

Parkinsonian movement problems can include slowness, stiffness, and difficulty initiating actions. Cognitive and mood changes may also occur, and clinical assessment examines these dimensions separately. A slow response on a timed task can reflect movement difficulty, cognitive processing, or contributions from both rather than a single assumed mechanism.

Differential diagnosis means comparing possible explanations for a person's symptoms. The clinician uses history, examination, appropriate testing, and follow-up to decide which conditions require attention. Favre's public disclosure provides a concrete reason to distinguish a player's reported diagnosis from speculation about an additional disease.



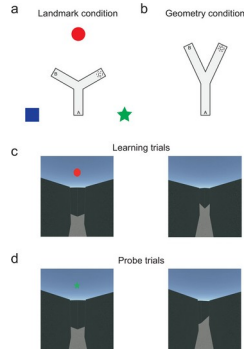
Photo: Brett Favre. Mass Communication Specialist 1st Class Jennifer A. Villalovos, USN. Cro; Public domain.

Navigation depends on the cues available to the person

A human navigation study compared the use of landmarks with the use of environmental geometry. Participants of different ages navigated a Y-shaped maze. Children and older participants had more difficulty using landmarks, while changing the geometry of the environment allowed them to navigate more effectively.

The original recording shows a participant undergoing disorientation in an immersive virtual setting. This procedure changes the information available about the starting direction before navigation. It helps separate remembering an environmental location from simply continuing in the direction the person was already facing before the trial.

The study found that apparent difficulty with world-centered navigation did not reflect a complete loss of that ability. Performance depended on whether the environment supplied useful landmarks or geometric information. In an everyday complaint about getting lost, the available cues and the person's strategy therefore need to be described.



Bécu et al. (2023), Fig. 1.



Human navigation: virtual disorientation. Bécu et al. (2023).

Play video

Jim McMahon described forgetting the purpose of an action

Jim McMahon publicly described memory difficulties in a 2011 report about former players' concussion concerns. One example was forgetting why he had entered a room. The report establishes that he described this experience; it does not supply a confirmed CTE tissue diagnosis or a complete neuropsychological assessment.

Remembering an intended action is called prospective memory. A person forms a goal and must retrieve it when the appropriate time or situation occurs. Interruptions can displace the goal from attention, while a distinctive environmental cue can help the person recover what they had intended to do.

An examiner asks how often such lapses occur, whether they represent a change, and whether they disrupt important activities. Forgetting an occasional minor intention differs from repeatedly missing medicines or appointments. The actions, frequency, and consequences make the complaint clinically interpretable without assigning its cause from the example alone.

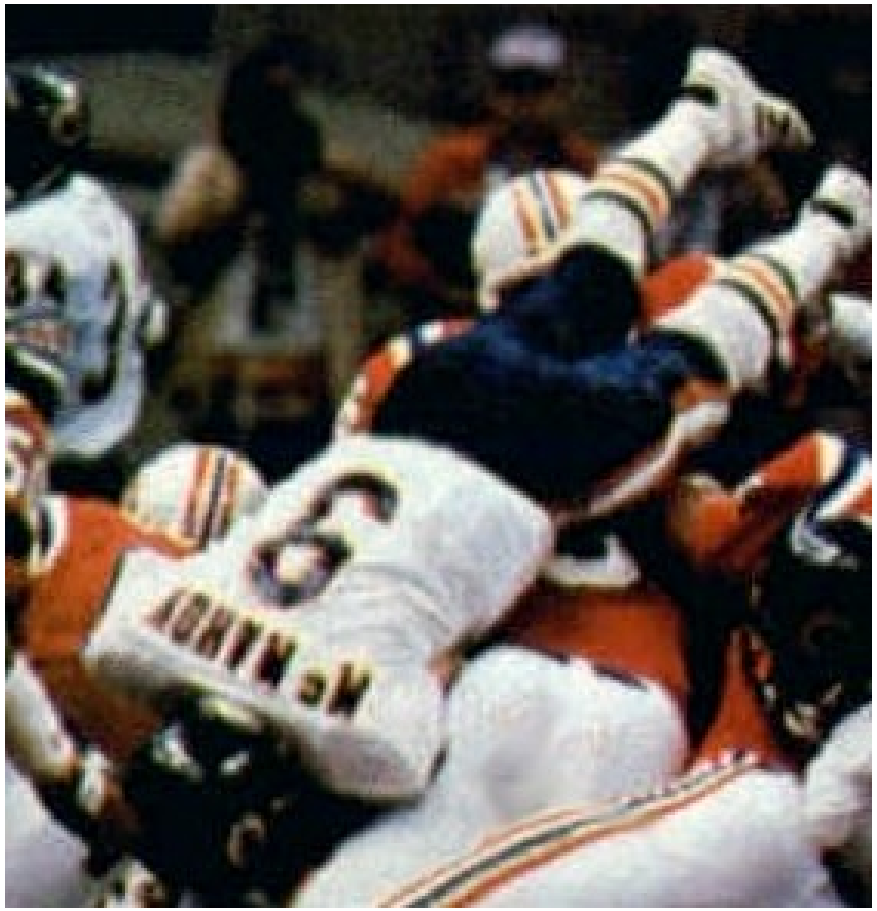


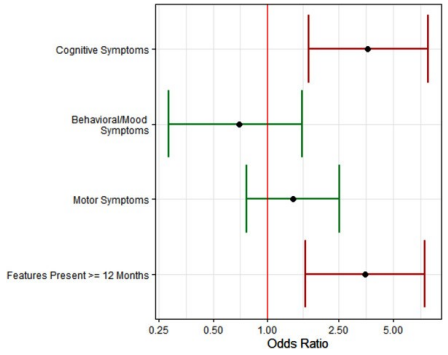
Photo: Jim McMahon. Jeno's; Public domain.

Older syndrome criteria captured nonspecific symptoms

Researchers tested the 2014 traumatic encephalopathy syndrome criteria against brain pathology in donors with repetitive-impact histories. Clinicians reviewed records and interviewed informants without knowing the tissue findings. Neuropathologists evaluated the brains without knowing the clinical classification, allowing the two assessments to be compared.

The older criteria identified many donors who had CTE but also identified many without it. Cognitive features were associated with the pathology more clearly than mood, behavioral, or motor symptoms in that analysis. Accompanying Alzheimer disease pathology further complicated the accuracy of the clinical classification.

The examination recording shows abstract-reasoning questions, including proverb interpretation. A person can answer some practical questions successfully yet interpret figurative language concretely. This separate clinical example demonstrates why an examiner records the pattern of responses rather than treating one behavioral description as a specific disease marker.



Mez et al. (2021), Fig. 2.



Clinical demonstration: abstract reasoning. Larsen and Stensaas (2020).

Play video

Football expertise differs from general cognitive testing

Joe Namath's quarterback career provides an example of expertise developed within a specialized activity. A quarterback learns formations, rehearses assignments, and recognizes familiar patterns of movement. Extensive knowledge can make a demanding sporting decision appear rapid because the person has practiced interpreting that particular kind of situation.

A standardized cognitive task removes much of that familiar context. It may require learning unfamiliar material, changing an arbitrary rule, or responding to items without help from teammates. Performance on these tasks is intended to measure particular abilities rather than knowledge of football strategy or a player's professional reputation.

The distinction also matters after retirement. A person may retain detailed knowledge of old plays while having difficulty learning new information or managing a new routine. An assessment therefore separates established knowledge from new learning and evaluates whether a change affects the activities the person currently needs to perform.

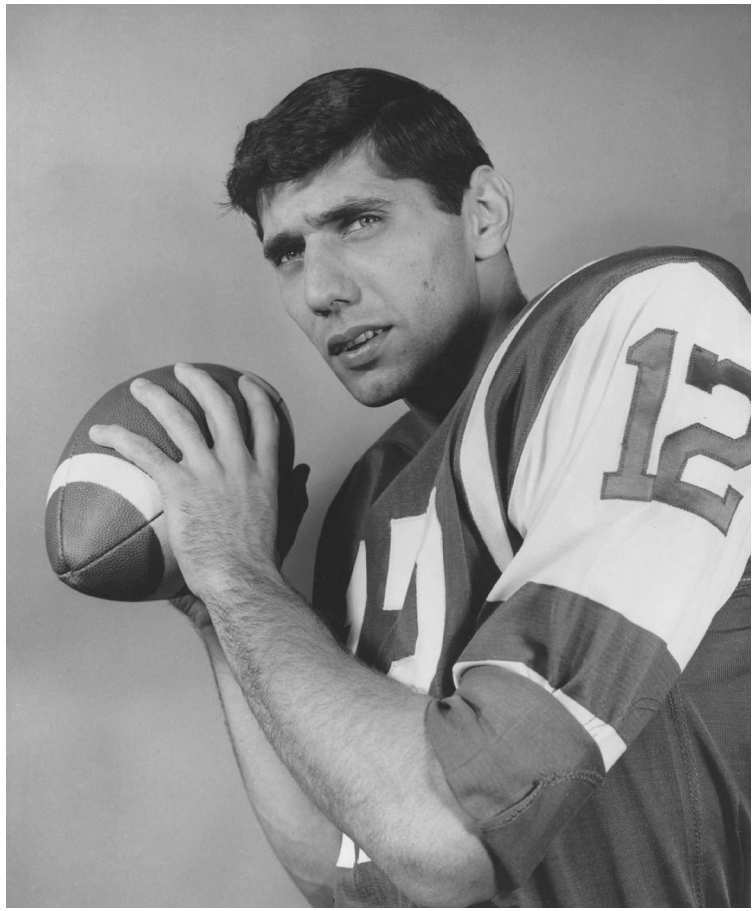


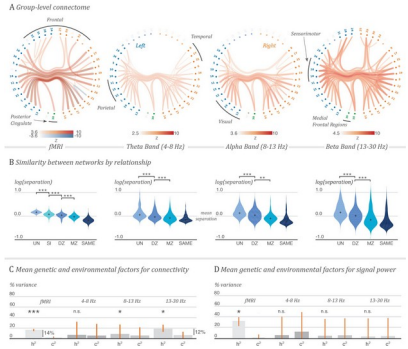
Photo: Joe Namath. New York Jets; Public domain.

Functional connectivity measures signal covariation

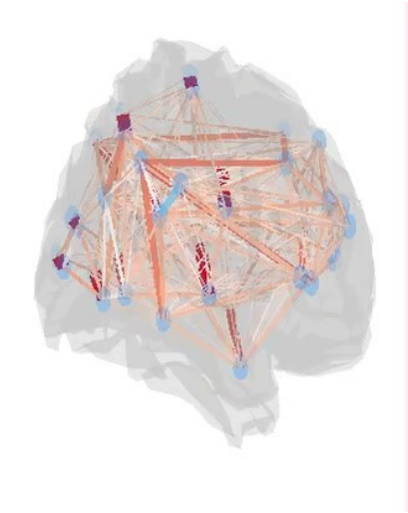
Researchers used functional MRI and magnetoencephalography to study how activity varies together across human brain regions. Functional connectivity means that two measured signals show a reproducible pattern of covariation. It does not, by itself, show a direct anatomical cable or establish that one region caused activity in the other.

The original fMRI visualization displays a network summarized from regional measurements. The study compared patterns among twins and other participants to estimate contributions from genetic and shared environmental factors. Genetic differences accounted for some variation in connectivity, while substantial variation remained beyond the shared-family environment examined here.

A brain network image therefore depends on the signal, analysis, and population used to construct it. This recording comes from a human connectivity study, not football-player CTE imaging. The same distinction is needed when reading injury studies that compare patterns of coordinated activity between groups.



Colclough et al. (2017), Fig. 1.



Human fMRI connectivity network. Colclough et al. (2017).

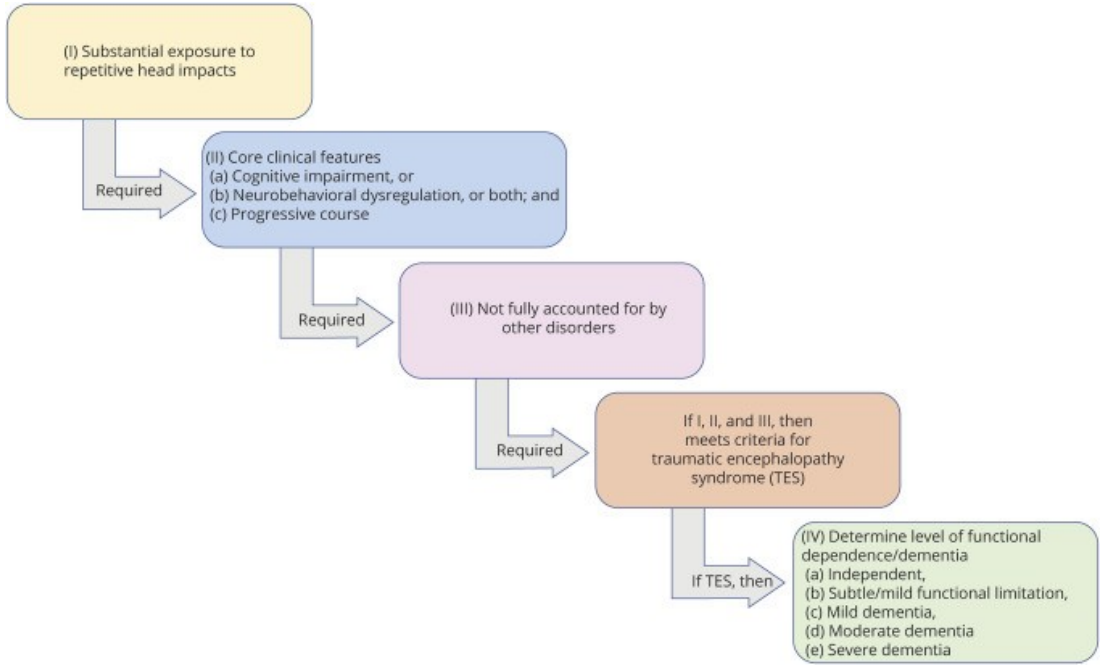
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A clinical history establishes change from earlier function

The 2021 research criteria for traumatic encephalopathy syndrome require substantial repetitive-impact exposure and a progressive clinical course. Core features involve episodic memory or executive impairment, neurobehavioral dysregulation, or both. The assessment also asks whether another neurological, psychiatric, or medical condition fully explains the clinical features.

Progressive change means that difficulties increase over time rather than remaining at the person's lifelong level. A history must therefore establish the earlier pattern of functioning. School records, work history, earlier testing, and observations from people who know the person can help distinguish longstanding differences from later decline.

The criteria separately grade functional dependence, from independence to severe dementia. Needing assistance with finances, medication, transportation, or basic personal care describes the consequences of impairment. These daily activities provide information that is not contained in a single low cognitive score or an isolated abnormality on a scan.



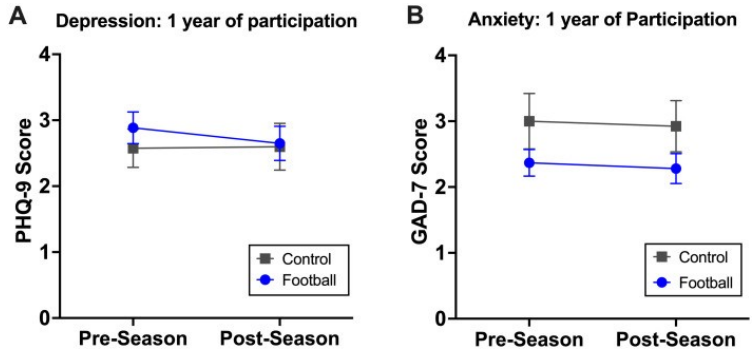
Katz DI et al. (2021), Fig. 1.

Youth studies did not show one uniform cognitive outcome

A longitudinal study followed youth tackle-football players across four seasons while measuring head-impact exposure and cognitive and behavioral outcomes. Few associations appeared between cumulative impacts and the measured outcomes. The associations that did appear were not consistent across the years or progressively stronger throughout follow-up.

This result differs from simply assuming that every exposure history must produce detectable decline during a short developmental interval. The study measured particular outcomes during childhood. Those results describe that cohort and observation period rather than resolving the long-term clinical course of every player who later reaches professional football.

The historical Princeton and Yale recording shows a much earlier form of the sport, including repeated plays and crowded contact. It supplies visible sporting context, not the exposure measurements from the youth study. Modern research requires its own impact recording, cognitive assessments, and follow-up rather than estimates based on this film.



Buddenbaum et al. (2025), Fig. 2.



Princeton–Yale football, 1903. Edison/Library of Congress (1903).

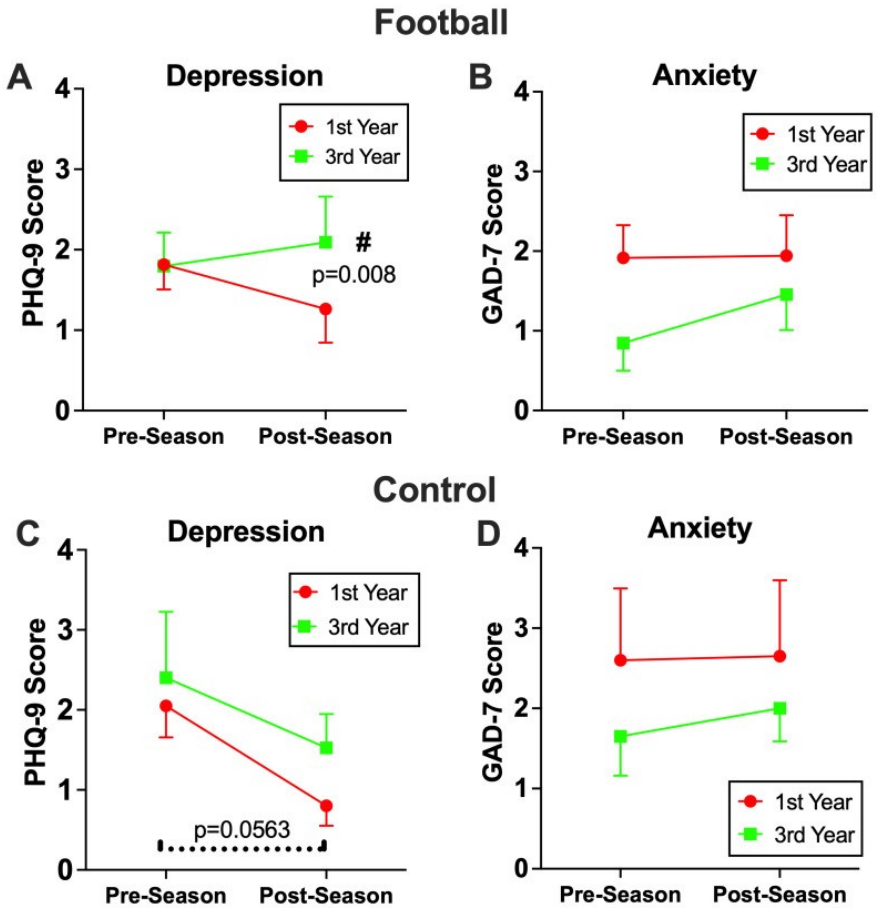
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Several seasons revealed a different symptom pattern

A prospective high-school study followed football and noncontact athletes over multiple seasons. Athletes completed depression and anxiety questionnaires before and after seasons, and football exposure was recorded with instrumented mouthguards. The researchers examined both changes within a season and changes across repeated years of participation.

Depression and anxiety scores were generally stable across the first one or two years in both sporting groups. Among participants followed for three consecutive years, the football group showed a different pattern in depressive symptoms across the observation period. The result depended on the duration of follow-up rather than a uniform immediate change.

A repeated questionnaire records symptom burden at the time it is completed. It is different from a neuropathological diagnosis or a demonstration that a specific molecular lesion caused the symptoms. Repeated observations allow investigators to describe when experiences change and compare those patterns with the measured history of participation.



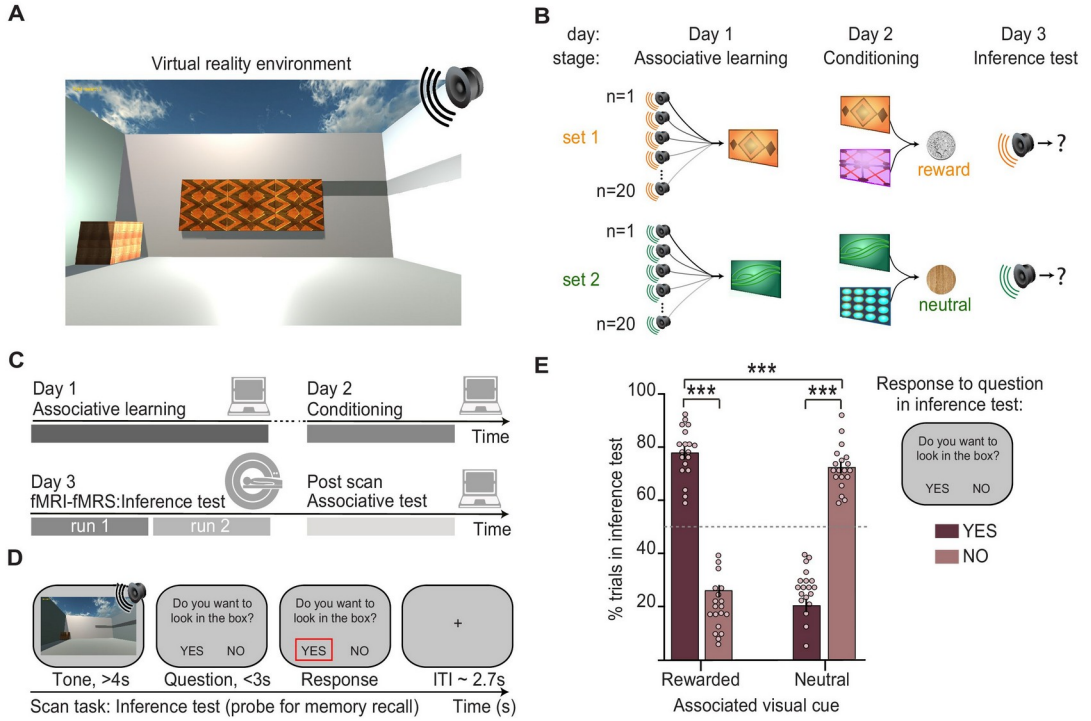
Buddenbaum et al. (2025), Fig. 4.

Remembering an event requires learning and later retrieval

Episodic memory allows a person to remember an event together with information about its context. Encoding is the process through which the experience is initially registered. Retrieval brings stored information back into use, and a retrieval cue provides part of the information that helps recover the rest of the event.

The hippocampus interacts with cortical regions that process the content of an experience. Remembering where an object appeared or what accompanied a sound can require reinstating a distributed pattern of information. Damage or altered activity can affect initial learning, retention over time, or the ability to recover the stored content.

A memory examination can therefore vary the learning opportunities, delay, and available cues. Free recall asks the person to produce information with relatively little help. Recognition supplies alternatives and asks which was encountered before. Comparing these conditions describes a pattern of performance more precisely than asking only whether memory is poor.



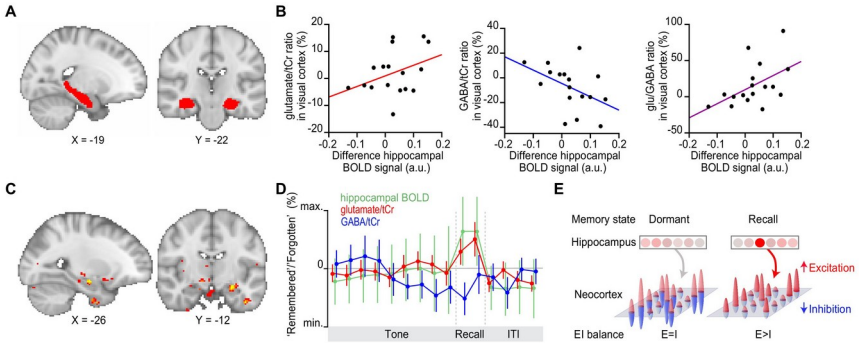
Koolschijn et al. (2021), Fig. 1.

Human recall changed activity and local chemical balance

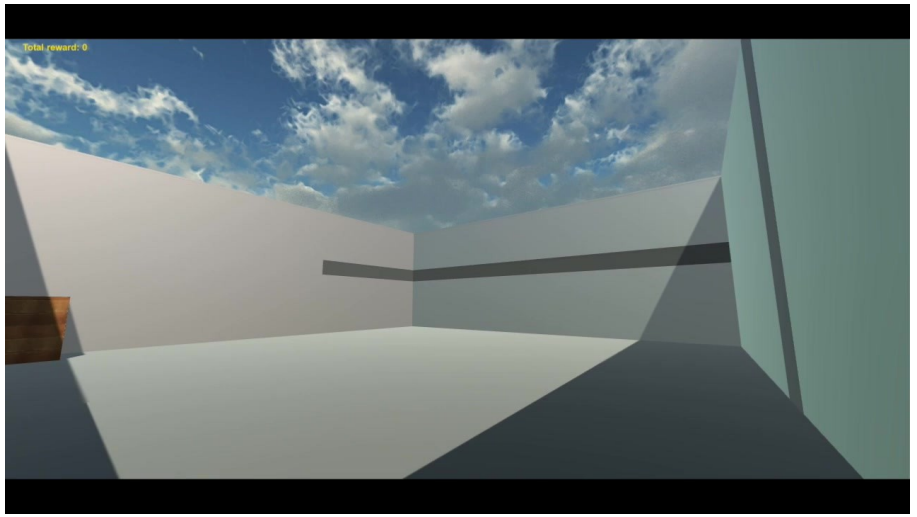
In a human memory experiment, participants learned associations involving sounds and visual information from a virtual environment. Researchers combined functional MRI with magnetic resonance spectroscopy, a technique that estimates concentrations of selected chemicals within a tissue volume. The experiment examined the interaction between hippocampal activity and visual-cortical memory reinstatement.

During recall, visual cortex showed a transient increase in the ratio of glutamate to GABA. These chemicals participate in excitatory and inhibitory signaling. Hippocampal activity predicted the fluctuations, consistent with a proposed mechanism in which hippocampal output helps make stored cortical information available for retrieval.

The original movie shows one of the visual trajectories presented with an auditory cue during the MRI inference task. It is a task stimulus rather than a movie of a participant's thoughts. The study measured ordinary human recall and supplies a mechanism for considering memory performance, not a validated CTE diagnostic procedure.



Koolschijn et al. (2021), Fig. 5.



Stimulus used during a human MRI memory task. Koolschijn et al. (2021).

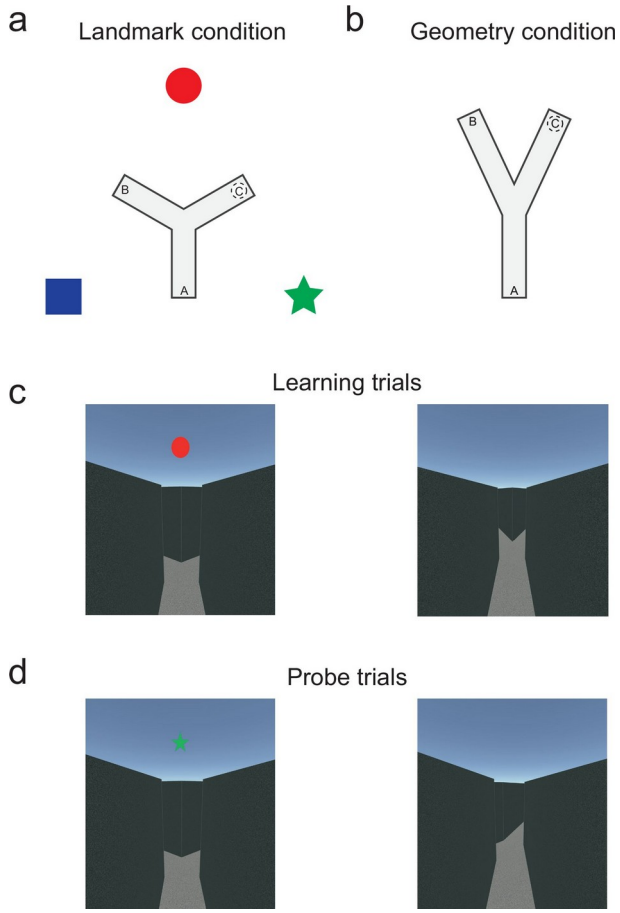
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A spatial complaint can involve several reference frames

An egocentric reference frame describes locations relative to the person's body, such as an object being on the left. An allocentric reference frame describes locations relative to the environment, such as a doorway being beside a particular landmark. Navigation can use both kinds of information while the person changes position.

Landmarks are recognizable features that help identify a location. Environmental geometry supplies information through the arrangement of boundaries and directions. A person can have difficulty recognizing a landmark while still using the shape of a corridor or room to orient successfully within the surrounding space.

Questions about getting lost should establish where the difficulty occurs and which cues are available. An unfamiliar intersection, a visually crowded setting, and a familiar route make different demands. The resulting description can guide testing of visual recognition, spatial memory, attention, and planning rather than assuming a single navigation faculty failed.



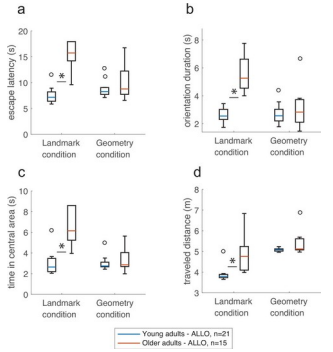
Bécu et al. (2023), Fig. 1.

Environmental geometry improved navigation across ages

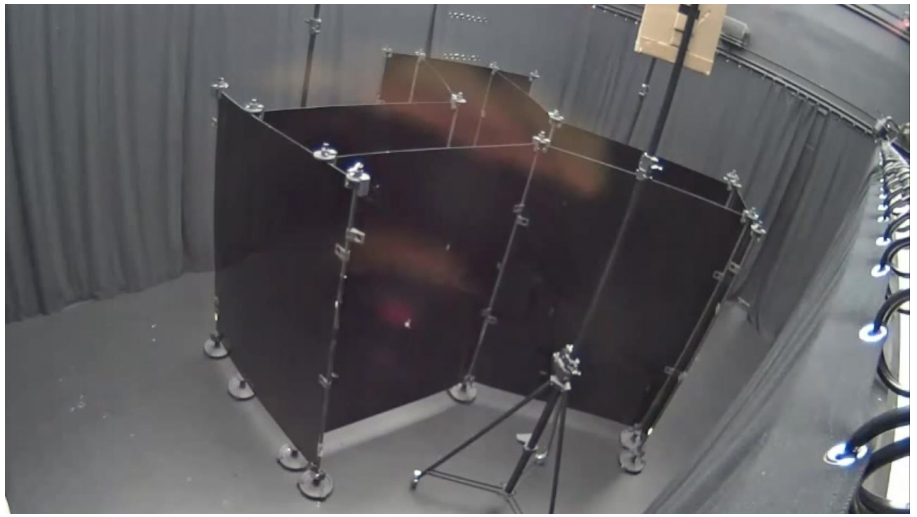
The navigation study tested children, younger adults, and older adults in environments that supplied different directional information. In one condition, the maze's geometry did little to distinguish directions, making landmarks especially important. In another, the environment had a geometric asymmetry that could guide orientation independently of particular landmarks.

Age differences were more apparent when successful navigation depended on landmarks. When geometry provided useful directional information, children and older participants could perform more like younger adults. The finding separated difficulty using one source of information from a general inability to represent a destination in the environment.

The original recording shows learning and probe trials in the real-world version of the task. Learning establishes a destination, while a probe tests whether the person can locate it under the specified conditions. These observable navigation actions help explain why changing a person's environment can change performance without changing the person's underlying diagnosis.



Bécu et al. (2023), Fig. 7.



Human navigation: learning and probe trials. Bécu et al. (2023).

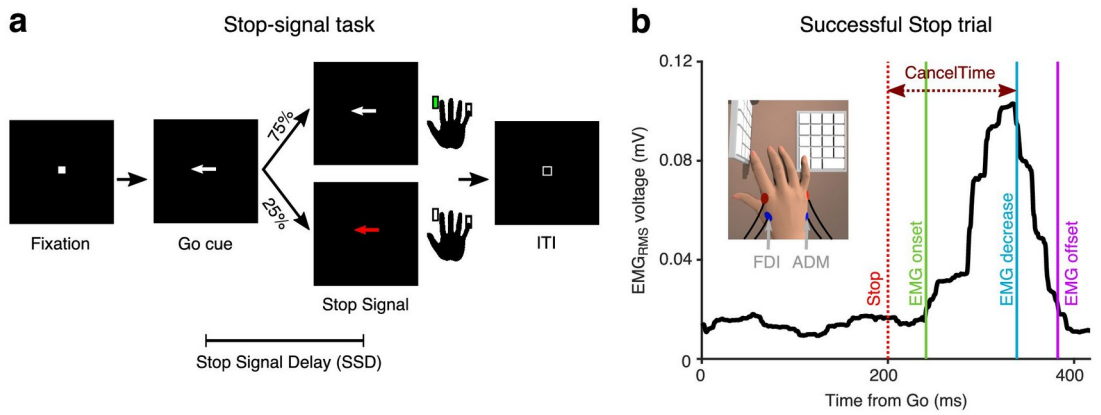
Play video

Executive control maintains goals and changes actions

Executive functions organize behavior around a goal. They include keeping a rule active, selecting a relevant response, suppressing an inappropriate action, and changing strategy when circumstances change. These operations depend on interacting cortical and subcortical systems rather than a single isolated center for good judgment.

Response inhibition is the ability to interrupt or prevent an action when the situation requires it. A player stopping before an illegal movement and a driver withholding a turn illustrate the demand. A laboratory stop task creates a controlled version by presenting an instruction to stop after a response has already been prepared.

An inhibition problem differs from forgetting the instruction or failing to perceive the signal. Researchers vary signal timing and record response initiation, muscle activity, and brain activity to distinguish these stages. The sequence of measurements helps locate where stopping becomes slower rather than describing every unsuccessful stop as impulsivity.



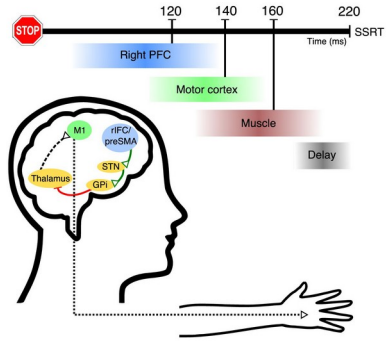
Jana et al. (2020), Fig. 1.

Stopping an action unfolded through measurable stages

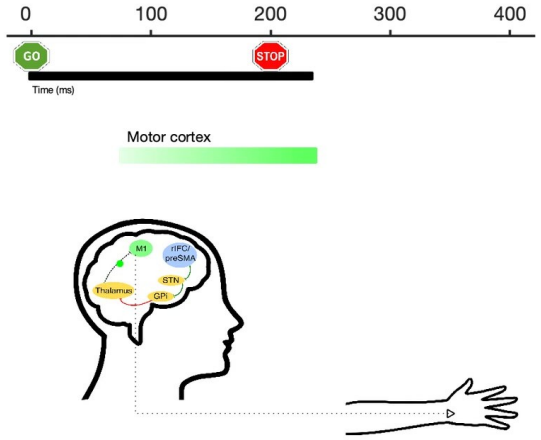
A human action-stopping study combined electrical recordings with measures of muscle activity and behavior. After the stop signal, frontal beta activity increased before a more general suppression of the motor system appeared. Suppression measured in the responding muscle followed, and the behavioral estimate of successful stopping occurred later.

The authors proposed a circuit in which frontal regions influence the subthalamic nucleus and basal ganglia output, reducing the thalamic drive that helps activate motor cortex. The original published animation displays this proposed sequence. It is an explanatory model based on the measurements, not direct footage of signals moving between individual brain structures.

The distinction between brain, muscle, and behavioral timing explains why one response time cannot identify every step in executive control. A slow or unsuccessful stop can involve delayed detection, circuit recruitment, or motor cancellation. These components provide more specific targets for measurement than a broad judgment about a person's character.



Jana et al. (2020), Fig. 6.



Published model of human action-stopping. Jana et al. (2020).

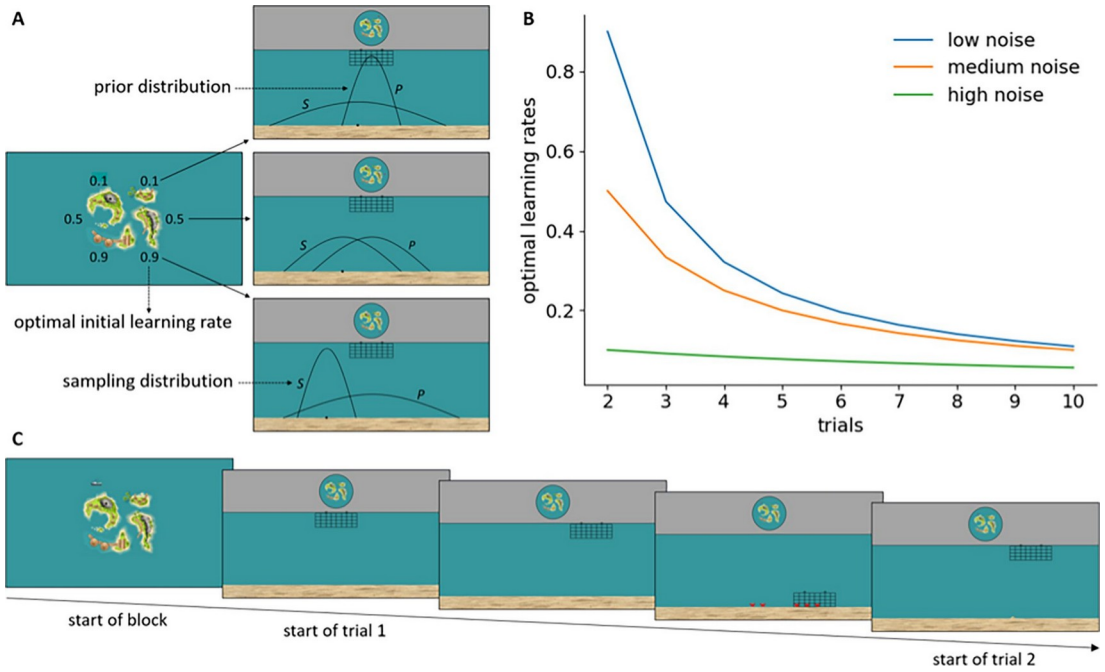
Play video

A learning rate controls updating after new evidence

A prediction is an expectation about what will happen. A prediction error is the difference between that expectation and the observed outcome. A learning rate determines how strongly a new outcome changes the next prediction, so the same surprising event can produce a small revision or a large one.

Rapid updating can be useful when the environment has genuinely changed. Slower updating can prevent a person from overreacting to a single unusual outcome in an otherwise stable setting. Flexible decision-making requires distinguishing a change in the underlying situation from random variation within the same situation.

A familiar sporting environment also contains expectations about opponents and likely outcomes. Cognitive experiments isolate updating by controlling how outcomes are generated. The resulting task measures how participants use new information, rather than assuming that persistence always reflects good strategy or that changing a response always reflects confusion.



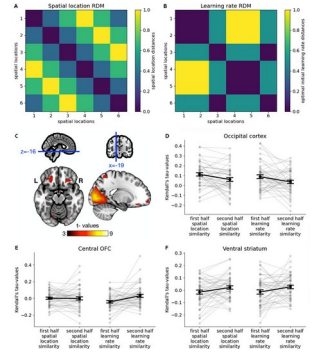
Simoens et al. (2026), Fig. 1.

People learned local and environment-specific rules

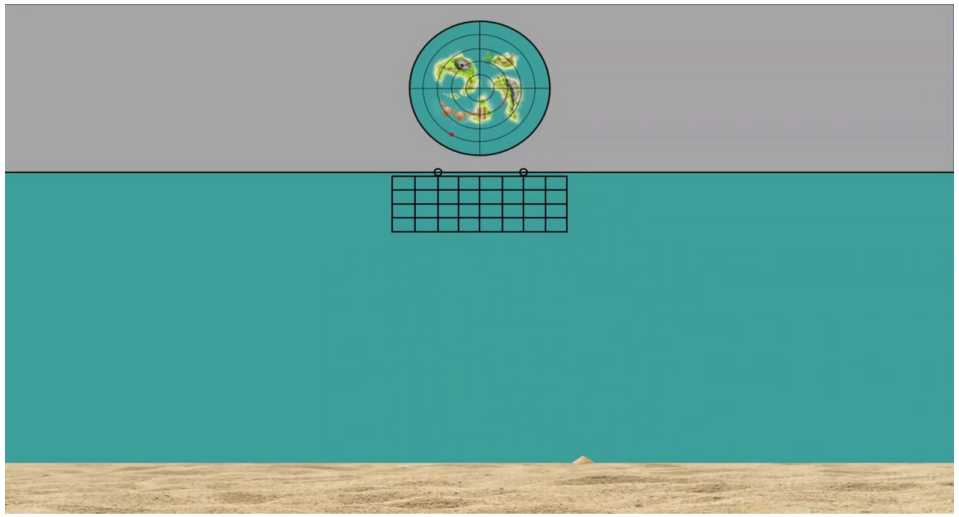
Participants in a human learning study estimated where to fish for crabs in several virtual island environments. The environments required different optimal learning rates. Researchers examined how participants changed their estimates after recent outcomes and whether they learned a more stable updating strategy for each island across repeated visits.

Participants adapted quickly within a block and also learned environment-specific differences that persisted when they returned to an island. Functional MRI patterns in central orbitofrontal cortex varied with these more global differences in learning rate. The study therefore separated immediate adjustment from slower learning about the kind of environment encountered.

The original movie demonstrates the crab-fishing task and the information available to the participant. It shows how a decision problem can make updating observable. The experiment involved human learning under controlled conditions; it did not establish that a particular learning-rate pattern identifies football-related injury or CTE.



Simoens et al. (2026), Fig. 6.



Human learning task: fishing for crabs. Simoens et al. (2026).

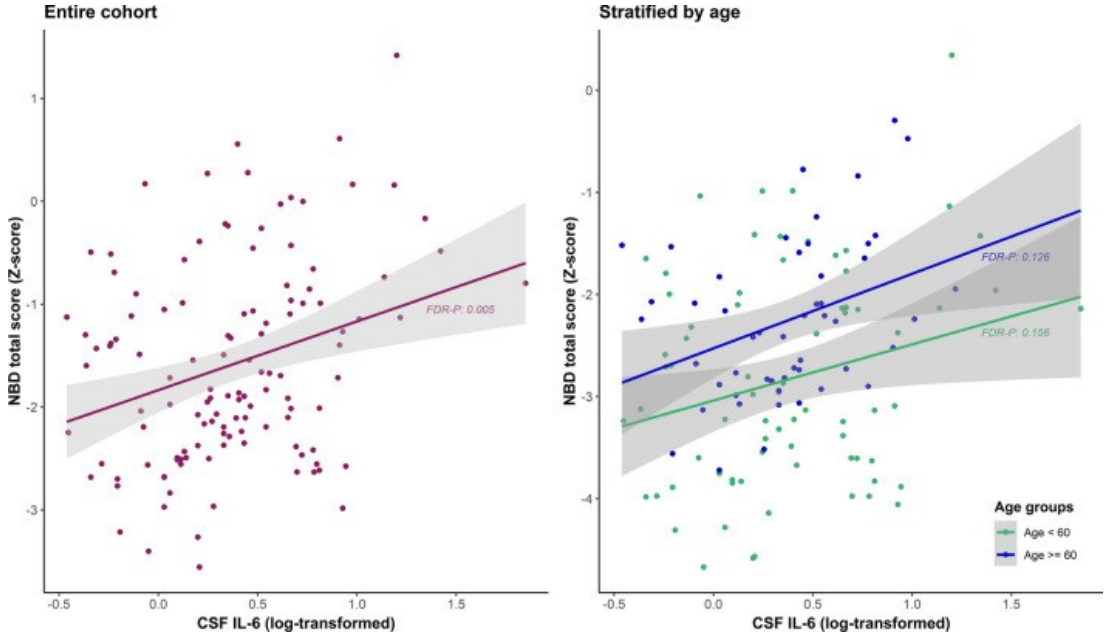
Play video

Emotional regulation changes the expression of a response

Emotional regulation refers to processes that influence the intensity, duration, and expression of an emotional response. A person may feel anger while still being able to pause, reconsider, and choose an appropriate action. Difficulty with regulation concerns how the response develops and is controlled rather than the mere presence of an emotion.

Affective lability describes rapid or marked shifts in emotional state. Emotional dyscontrol describes difficulty modulating the expression of emotion. Impulsivity concerns acting with insufficient consideration of consequences. These features can coexist, but careful assessment asks which actions occur and under what circumstances instead of treating the terms as synonyms.

The traumatic encephalopathy syndrome research criteria include a specified pattern of neurobehavioral dysregulation. A clinical history also considers psychiatric, neurological, and medical explanations for the behavior. The task is to describe new changes and their functional consequences while examining whether another condition adequately accounts for them.



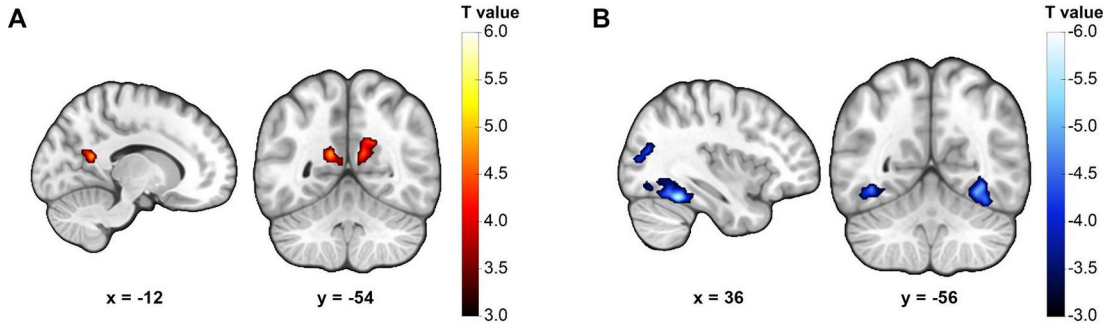
van Amerongen et al. (2024), Fig. 1.

Social judgments can be represented as a cognitive space

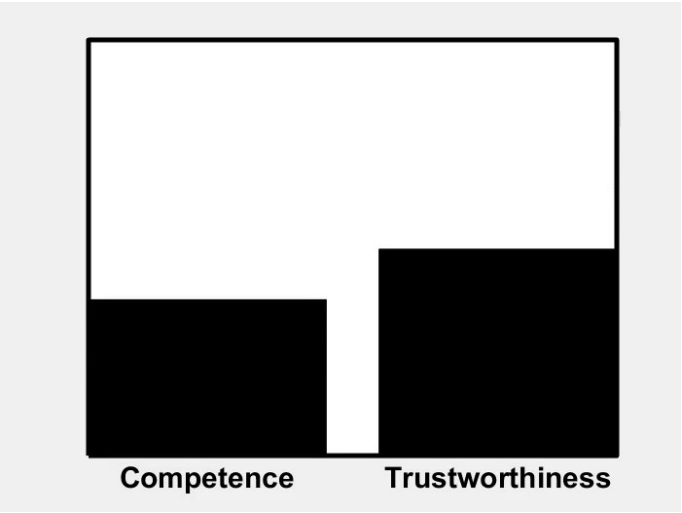
A human imaging study organized information about other people along dimensions of competence and warmth. Participants learned positions in this abstract social space and then navigated toward a learned target. The task tested whether a structured representation could organize social information even when there was no physical location to visit.

Brain activity represented distance in regions including the precuneus and visual-association cortex. The authors also found partial evidence of grid-like response patterns in medial prefrontal and entorhinal regions. The findings concerned how a controlled social representation was organized, rather than a direct measurement of everyday empathy or interpersonal success.

The original movie shows the morphing stage used during the recall task. It makes the task's changing social information visible. When considering social difficulties after injury, the relevant cognitive operations can include learning information about others, retrieving it, and using it to guide a judgment, each of which requires its own measurement.



Liang et al. (2024), Fig. 2.



Recall task in an abstract social space. Liang et al. (2024).

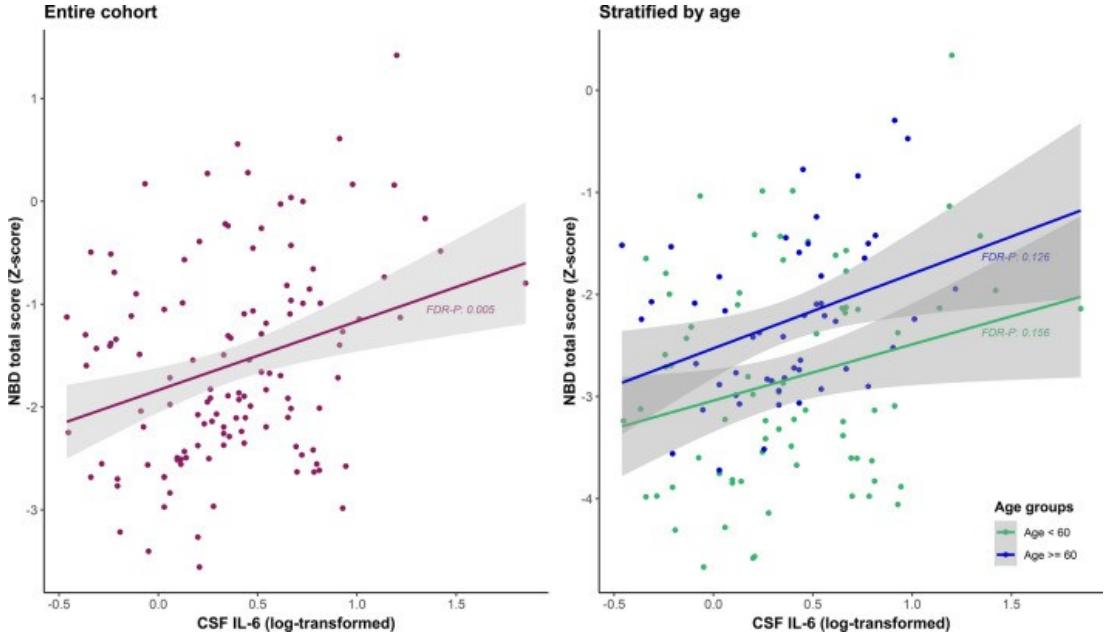
Play video

Behavioral dysregulation differs from a mood diagnosis

A depressive disorder concerns a sustained pattern of mood and associated symptoms, such as reduced interest, disrupted sleep, or changes in energy. Neurobehavioral dysregulation instead describes difficulty controlling emotion and behavior. A person can experience either pattern or both, so one label should not automatically replace the other.

Researchers studying former players separate components such as explosivity, impulsivity, affective lability, and emotional dyscontrol. Questionnaires and interviews ask about characteristic experiences and actions. The pattern across responses can be summarized into dimensions, allowing investigators to examine whether different biological measures accompany particular kinds of difficulty.

A biological association with one symptom dimension does not imply an identical association with every psychiatric diagnosis. The fluid sampled also matters because cerebrospinal fluid and blood are different compartments. A careful account identifies the measured protein, the sampled compartment, and the specific psychological feature that varied with it.



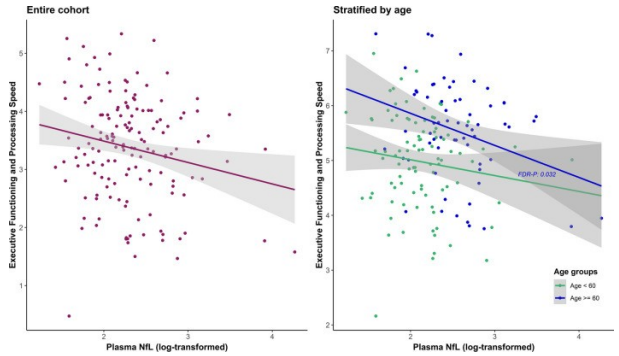
van Amerongen et al. (2024), Fig. 1.

Inflammatory measures varied with behavioral features

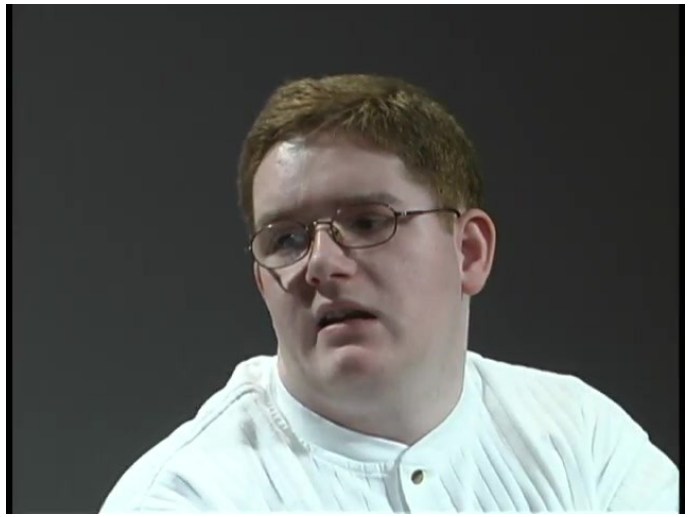
In the DIAGNOSE CTE study, former players with neurobehavioral dysregulation had higher cerebrospinal-fluid interleukin-6 than players without that classification. Interleukin-6 is an immune-signaling protein. Higher concentrations also accompanied greater emotional dyscontrol, affective lability, impulsivity, and an overall dysregulation measure within the observational analyses.

Among older players, blood neurofilament light was associated with emotional dyscontrol and impulsivity as well as poorer executive performance and processing speed. Neurofilament light is a structural neuronal protein that can enter body fluids after injury. These measures describe different biological processes and were not interchangeable readings of a single mechanism.

The separate clinical recording demonstrates difficulty maintaining attention and manipulating a sequence backward. It shows an observable executive demand rather than the behavior of a study participant. The paper's fluid associations and a person's task performance occupy different levels of assessment, even when both concern difficulties relevant to clinical functioning.



van Amerongen et al. (2024), Fig. 3.



Clinical demonstration: attention difficulty. Larsen and Stensaas (2020).

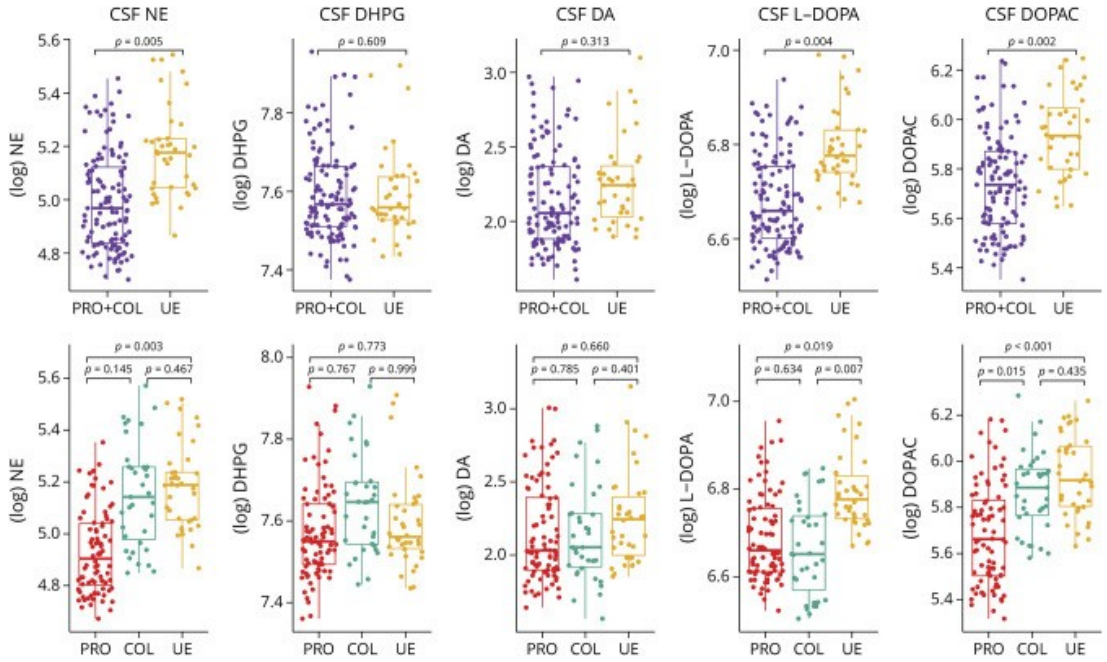
Play video

CSF chemistry did not provide a simple explanation

Researchers measured catecholamines and related compounds in cerebrospinal fluid from former football players and unexposed individuals. Catecholamines include dopamine and norepinephrine, which modulate neuronal signaling. The study measured concentrations and metabolites in fluid rather than directly recording transmitter release from particular circuits during a task.

Former players showed lower norepinephrine, a dopamine precursor, and a dopamine metabolite on selected comparisons. These group differences did not separate the clinical CTE-probability categories or parkinsonism categories examined in the study. A difference between exposed and unexposed groups therefore did not become a specific clinical classification marker.

Additional analyses within the college-player subgroup found associations between some compounds and behavioral dysregulation, anxiety, and executive performance. The restricted subgroup and cross-sectional sampling define what was observed. The results do not imply that every memory or mood complaint can be explained by a single low transmitter concentration.



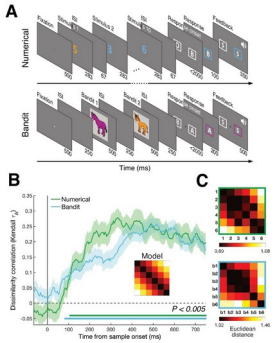
van Amerongen et al. (2025), Fig. 2.

New values can borrow the structure of familiar quantities

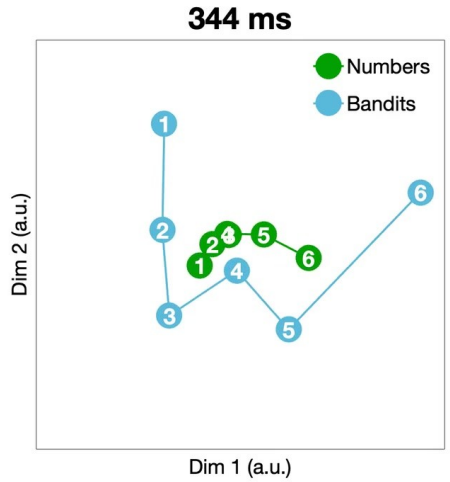
In a human reward-learning experiment, participants first viewed symbolic numbers and later learned about the reward probabilities of unfamiliar visual items. Researchers compared the neural representations evoked during the two tasks. The question was whether newly learned values could acquire a structure resembling an already familiar representation of numerical magnitude.

Items associated with better outcomes came to resemble larger numbers in the measured representation, while poorer items resembled smaller numbers. Neural-network simulations examined how this alignment could support learning and transfer. The original movie visualizes the geometry of the representations rather than recording a participant physically moving through a space.

Value learning differs from remembering a factual event or suppressing a movement. A person may understand a task's instruction yet update the value of an option differently after outcomes. Cognitive assessment can separate these operations by controlling the choices, feedback, and information available at each stage of the decision.



Luyckx et al. (2019), Fig. 1.



Neural representations of number and reward. Luyckx et al. (2019).

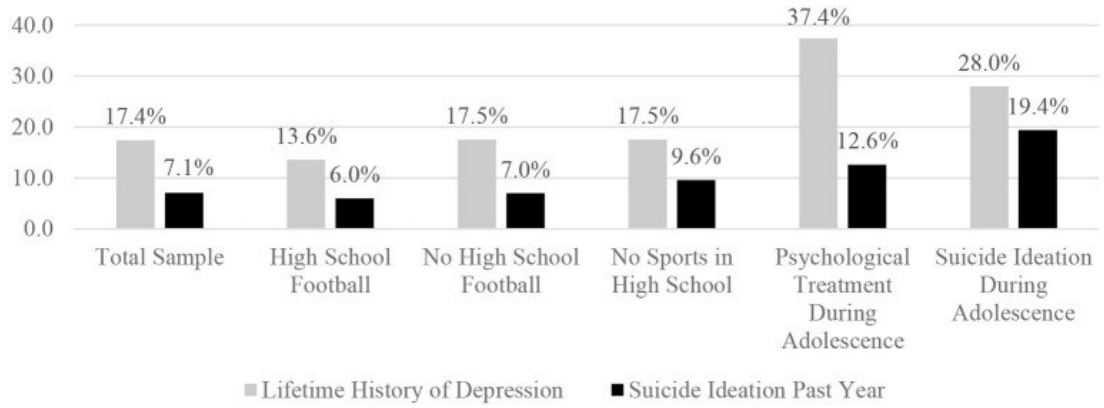
Play video

One national study found no adult mood-risk increase

A national longitudinal study compared adult mental-health outcomes in men who had reported playing high-school football with those who had not. The investigators used information collected during adolescence and again in adulthood. They examined depression, anxiety or panic diagnoses, recent depressed mood, counseling, and suicidal thoughts as separate outcomes.

The football and comparison groups did not differ on the measured adult outcomes in that analysis. Mental-health difficulties reported during adolescence, however, predicted greater difficulties more than two decades later. The individual's earlier psychological history therefore contained information about later outcomes that was not captured by football participation alone.

This finding concerns high-school participation and the follow-up period studied, rather than the exposure of every professional player. It also differs from estimating CTE prevalence in donated brains. The cohort design follows people selected through a population study instead of beginning with families who chose to donate tissue after a person's death.



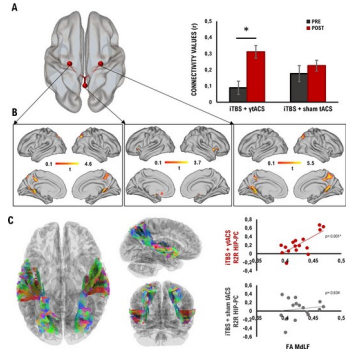
Iverson et al. (2021), Fig. 1.

Stimulation changed associative memory in volunteers

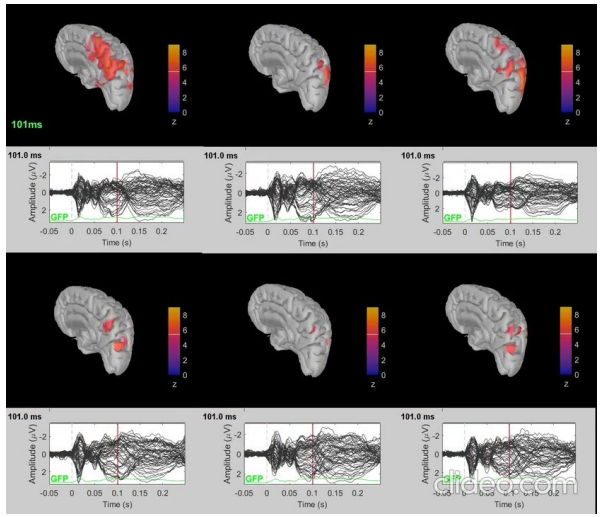
A human study applied two forms of noninvasive stimulation over the precuneus, a cortical region that interacts with memory networks. Intermittent theta-burst magnetic stimulation was paired with gamma-frequency alternating current. The investigators compared the combined procedure with magnetic stimulation alone and with a control condition that omitted active alternating current.

Only the combined procedure improved associative memory in the reported comparisons. The study also observed changes in gamma activity and precuneus–hippocampus functional connectivity. The original TMS–EEG visualization displays reconstructed responses before and after stimulation, making the measured electrophysiological response distinct from the later behavioral memory score.

These experiments involved healthy volunteers rather than a treatment trial in people with confirmed CTE. They test whether a specified intervention can alter human memory and associated signals under controlled conditions. Establishing clinical benefit after football-related injury would additionally require appropriate patients, meaningful functional outcomes, and comparison with an adequate control procedure.



Borghi et al. (2025), Fig. 4.



TMS–EEG responses around stimulation. Borghi et al. (2025).

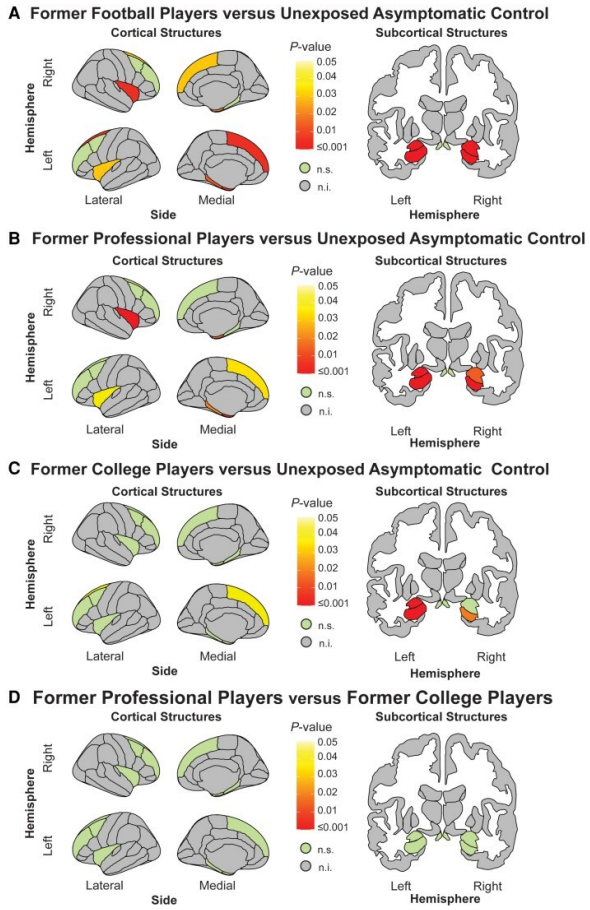
Play video

MRI measures tissue contrast and anatomy

Magnetic resonance imaging, or MRI, uses magnetic fields and radiofrequency signals to measure properties of tissue. Structural sequences produce contrasts that help distinguish gray matter, white matter, and fluid. The resulting image is an indirect measurement reconstructed from signals, rather than a photograph of thoughts, memories, or personality.

A structural analysis can estimate the volume of the hippocampus or the thickness of the cerebral cortex. The hippocampus participates in episodic memory, while cortical regions contribute to many interacting operations. An anatomical difference can motivate a question about function, but a person's performance still requires direct behavioral assessment.

Images also depend on acquisition settings, movement, and the method used to define anatomical boundaries. Group studies use consistent procedures to compare participants. Clinical interpretation considers the image alongside history and examination rather than assigning a psychological symptom to whichever area appears different on a colorful display.



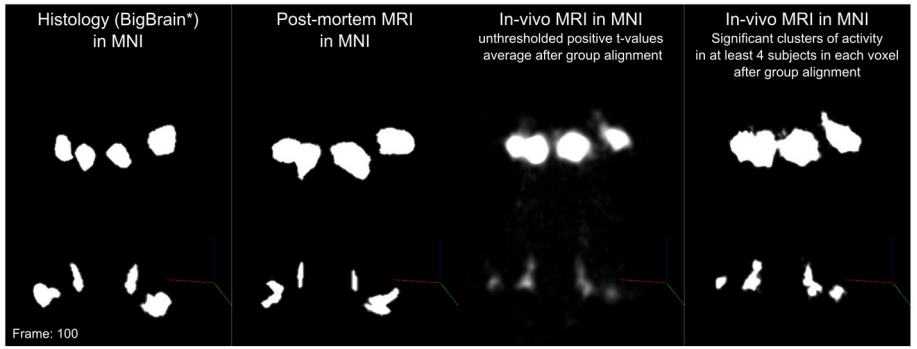
Arciniega et al. (2024), Fig. 2.

Human MRI maps can be checked against detailed anatomy

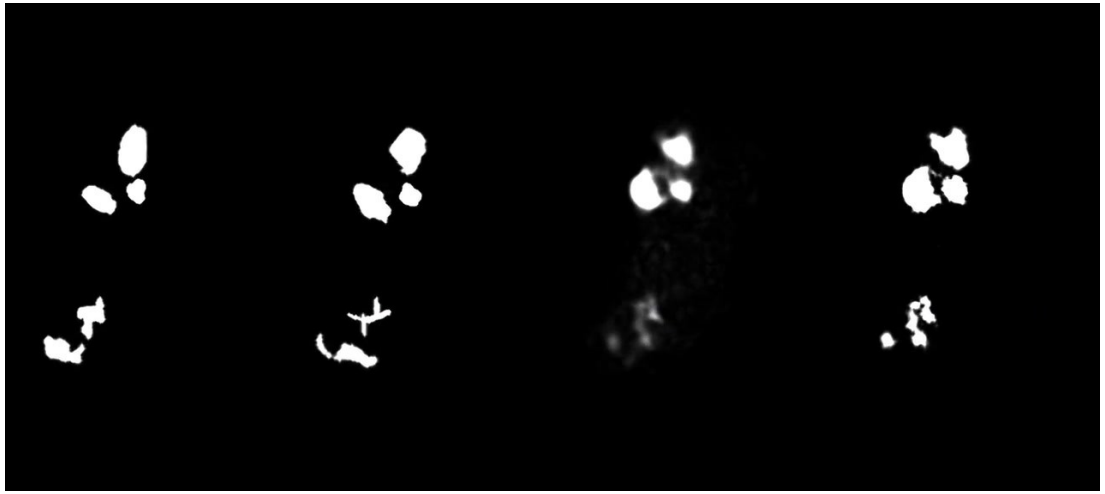
Researchers mapped the human subcortical auditory system using histology, postmortem MRI, and living-participant MRI at high magnetic field strength. The relevant structures are small and close together. Combining methods allowed the investigators to compare locations and pathways across preparations with very different spatial resolution.

The original reconstruction compares three-dimensional volumes after alignment into a common reference space. Histological detail and MRI contrast do not provide identical information, so agreement and displacement must be measured. A reference coordinate system helps compare studies, but registration does not make each person's anatomy identical to the reference.

This study supplies an example of validating an imaging map against anatomical information. Its participants and tissue were not a football CTE cohort. The same measurement issue arises when a player study labels a small structure or interprets a regional difference: the region must first be located and measured reliably.



Sitek et al. (2019), Fig. 9.



Human anatomical volumes in reference space. Sitek et al. (2019).

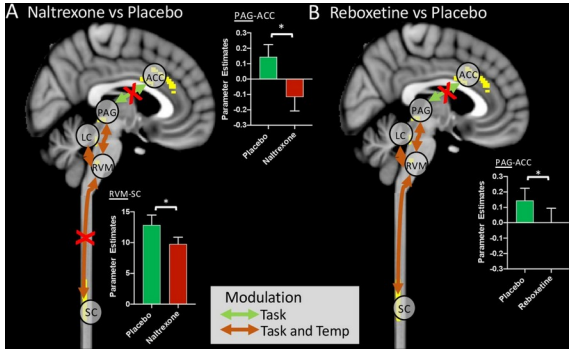
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Attention changed descending pain control

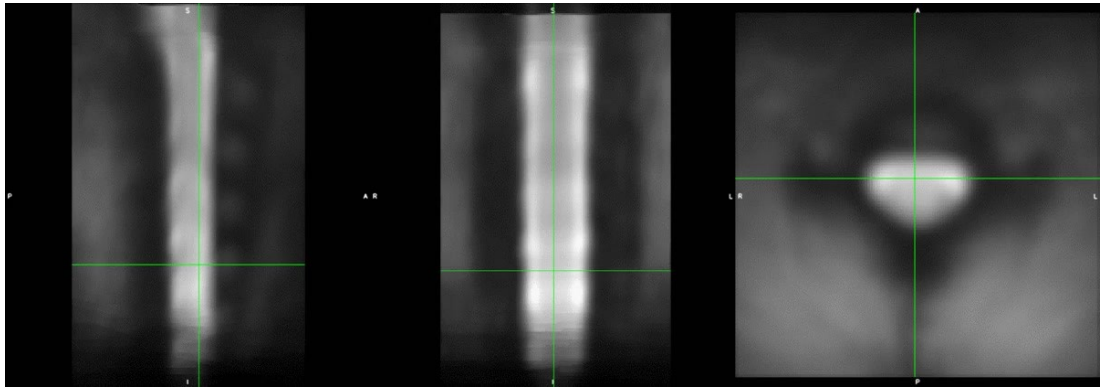
A human experiment combined an attention task with painful heat while recording functional MRI from the brain, brainstem, and spinal cord. Attention reduced reported pain under the studied conditions. Researchers examined descending pathways through which cortical activity could alter the processing of incoming signals rather than treating pain as a direct reading of tissue damage.

Blocking opioid receptors with naltrexone impaired the attentional reduction of pain and altered connectivity within the proposed control system. Increasing noradrenergic signaling with reboxetine did not produce the same change in attentional analgesia. The drug comparisons tested which signaling process contributed to the measured effect in that task.

The original recording demonstrates alignment of functional spinal images with an anatomical template. It shows a registration step used in the experiment, not a scan of a former player. Pain, attention, and cognitive performance can influence one another, making current pain an important part of the clinical history when evaluating everyday mental functioning.



Oliva et al. (2022), Fig. 5.



Functional and anatomical spinal MRI alignment. Oliva et al. (2022).

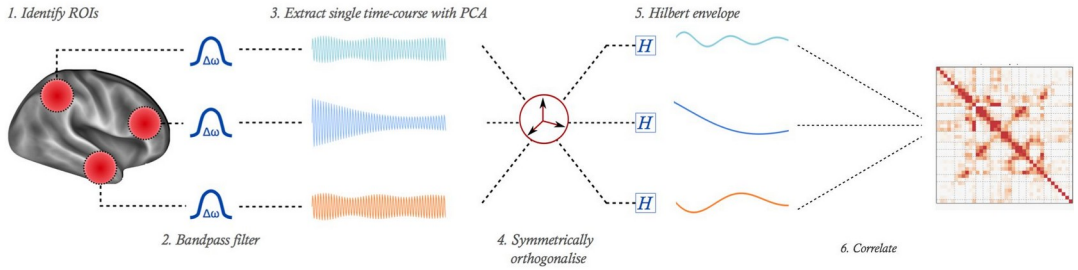
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MEG and fMRI measure different features of a network

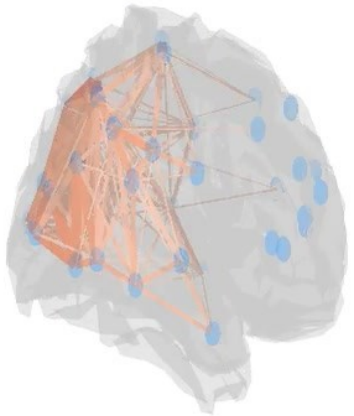
The human connectivity study measured networks with both functional MRI and magnetoencephalography, or MEG. MEG detects magnetic fields associated with neuronal electrical activity. Functional MRI measures a slower blood-oxygenation response, so the two methods describe activity through different physical signals and different time scales.

The selected original visualization shows connectivity derived from alpha-band activity rather than the fMRI network shown earlier. Alpha refers to a range of electrical oscillation frequencies. The study compared how connectivity varied among relatives across several signal types, finding that the measured contributions depended in part on the modality and frequency band.

A connection on such a display is defined by the analysis applied to recorded signals. It does not necessarily represent direct communication through one fiber bundle. When a study reports altered connectivity after injury, interpretation requires identifying the modality, frequency range, and specific measure that produced the network difference.



Colclough et al. (2017), Fig. 3.



Human MEG alpha-band connectivity. Colclough et al. (2017).

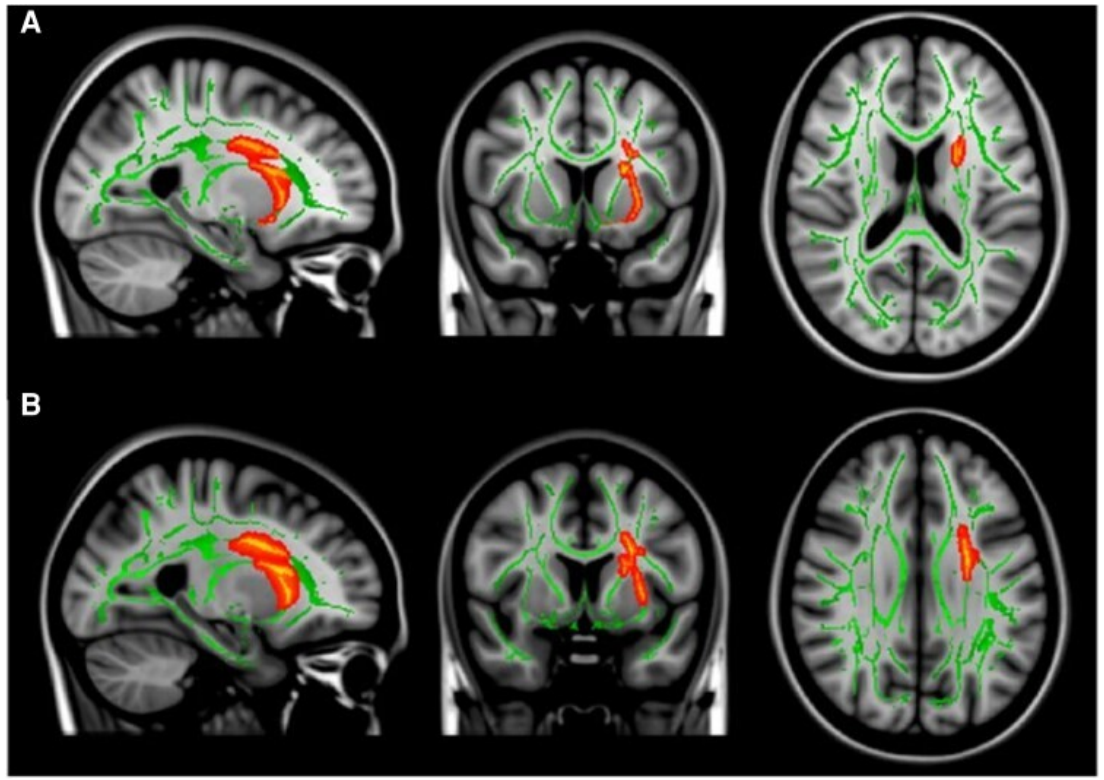
Play video

Diffusion imaging measures water movement in tissue

Diffusion MRI measures how water moves within tissue over a short interval. In white matter, cell membranes and tissue organization can make movement differ across directions. Diffusion tensor imaging summarizes this directional pattern using a simplified mathematical description at each measured location in the brain.

Fractional anisotropy describes how strongly the estimated diffusion differs across directions. It is not a direct count of axons or a universal scale of healthy white matter. Fiber orientation, crossing pathways, fluid compartments, and tissue changes can each influence the value, so either an increase or decrease requires interpretation.

Tractography uses local directional estimates to reconstruct possible routes through the image volume. These streamlines are computational estimates derived from the measurements. They do not directly show individual axons, the direction in which a neuron sends impulses, or a complete inventory of connections between the regions displayed.



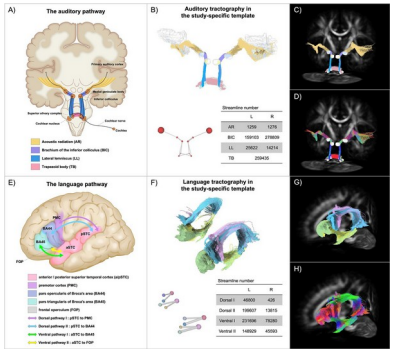
Arciniega et al. (2026), Fig. 1.

A reconstructed pathway depends on the imaging model

Researchers studying children with profound hearing loss combined structural and diffusion imaging to evaluate auditory and language pathways. They used directional tissue information to compare network development across different peripheral anatomical conditions. The analysis examined pathways that could not be adequately evaluated from an ordinary visual inspection of the ear alone.

The original movie displays a three-dimensional reconstruction of the language pathway. Its displayed bundles and regions are reconstructed from imaging measurements rather than photographed individual fibers. The paper distinguished auditory-pathway and language-pathway findings and examined how anatomical measures related to outcomes after hearing interventions in the studied children.

This developmental study is separate from football research. Its reconstruction makes the method used in white-matter studies visible and shows why an estimated tract is different from a cognitive score. A player study using tractography must separately establish which measured pathway property differs and whether that property predicts a clinically meaningful outcome.



Wang et al. (2023), Fig. 2.



Published diffusion-based language pathway. Wang et al. (2023).

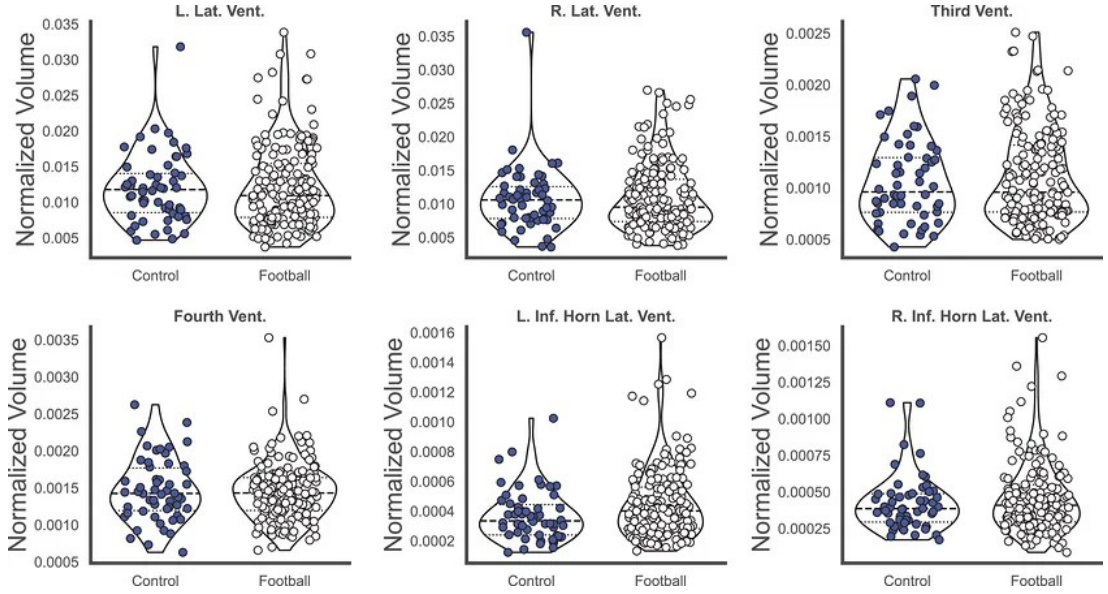
Play video

Ventricular enlargement describes a change in fluid spaces

The ventricles are connected fluid-filled spaces inside the brain. The lateral ventricles extend into each cerebral hemisphere, and their inferior horns lie near medial temporal structures. MRI can measure these spaces because fluid produces a different signal from surrounding tissue under appropriate acquisition settings.

A larger fluid space can accompany changes in surrounding brain tissue, but volume alone does not identify the biological cause. Development, aging, pressure-related processes, and neurodegenerative disorders can influence ventricular appearance. The particular region and the pattern across the rest of the image help determine which explanations require consideration.

A study can ask whether ventricular dimensions vary with estimated exposure to repetitive impacts. That is different from asking whether the same dimensions distinguish cognitive impairment or predict tissue-confirmed CTE. The imaging measure, exposure estimate, and clinical outcome are separate variables that need to be compared explicitly.



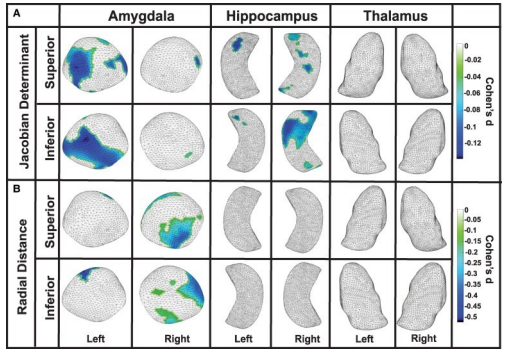
Szekely et al. (2026), Fig. 1.

Local brain shape differed in former football players

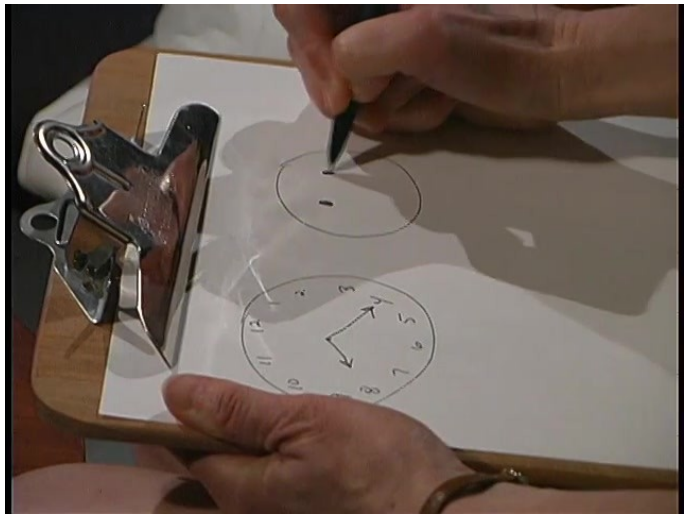
A DIAGNOSE CTE analysis examined the shapes of the hippocampus, amygdala, and thalamus in former football players and unexposed controls. The hippocampus participates in memory, the amygdala in learning about significant events, and the thalamus in several sensory and cognitive circuits. Shape analysis tested local differences beyond total regional volume.

Former players showed surface contractions in parts of the hippocampus and amygdala, with additional associations involving age and exposure estimates. The analysis did not find the same structural separation between players with and without the traumatic encephalopathy syndrome classification. Anatomical differences between exposure groups therefore answered a different question from clinical classification.

The separate examination recording shows visuospatial construction, including drawing tasks. Such tasks require perceiving spatial relations, maintaining an instruction, and organizing a response. They demonstrate one kind of function a clinician may examine; the recording is not from the imaging cohort and does not reveal the pictured person's hippocampal shape.



John et al. (2025), Fig. 1.



Clinical demonstration: visuospatial construction. Larsen and Stensaas (2020).

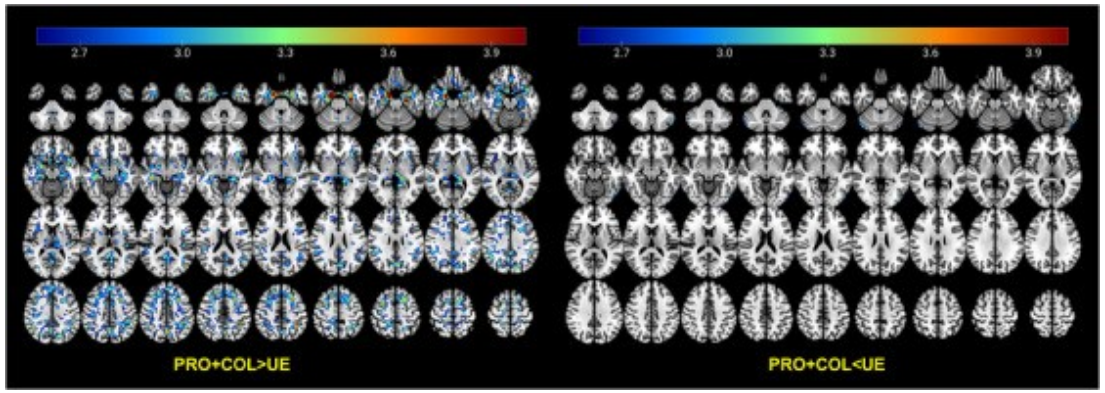
Play video

PET records the distribution of a radioactive tracer

Positron emission tomography, or PET, measures radiation emitted after a labeled compound is administered. A tracer is selected for a property such as binding to a molecular target. The scanner reconstructs where the signal is detected, and the analysis estimates tracer uptake under specified conditions and assumptions.

A tracer described as tau-sensitive must still be tested against the form and distribution of tau present in the disease being investigated. Different tau assemblies can interact differently with a compound. Uptake can also reflect binding outside the intended target, so a bright area does not automatically identify a particular pathological lesion.

An uptake ratio compares a region's signal with a reference measurement. Reference-region choice, timing after administration, and image processing can influence the result. A PET study must therefore establish how the measured signal corresponds to tissue pathology before it can support a diagnosis in an individual living player.



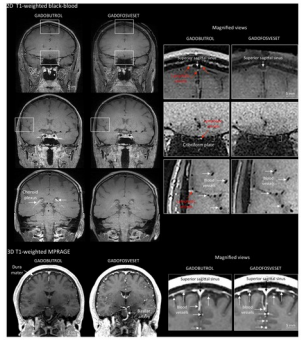
Su et al. (2024), Fig. 2.

MRI contrast made meningeal structures visible

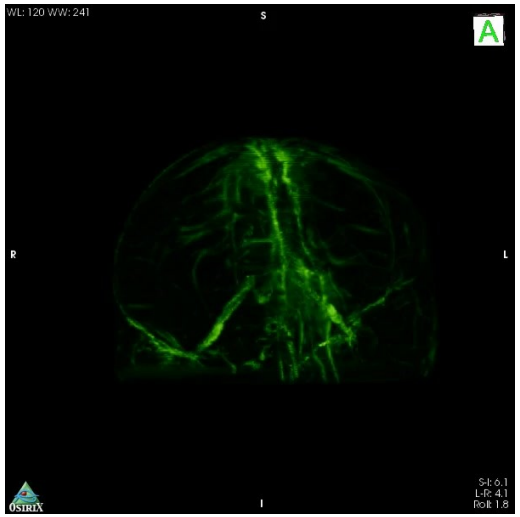
A human and nonhuman-primate study used specialized MRI sequences and contrast agents to investigate vessels in the meninges, the tissues surrounding the brain. The investigators compared two agents with different movement through vascular boundaries. Contrast behavior helped distinguish structures that were not easily visible on ordinary images.

The original human reconstruction shows enhancing structures alongside the dural venous sinuses. The study combined imaging with anatomical and molecular evidence to identify meningeal lymphatic vessels. It provides an example in which a visible structure depended on the acquisition sequence and the properties of the administered compound.

This is contrast-enhanced MRI, not tau PET and not an image of a player's psychological state. A scan's colors and three-dimensional form can look persuasive while representing very different measurements. The relevant interpretation begins with what generated the signal and which independent observations established the identity of the structure shown.



Absinta et al. (2017), Fig. 2.



Human meningeal structures on contrast MRI. Absinta et al. (2017).

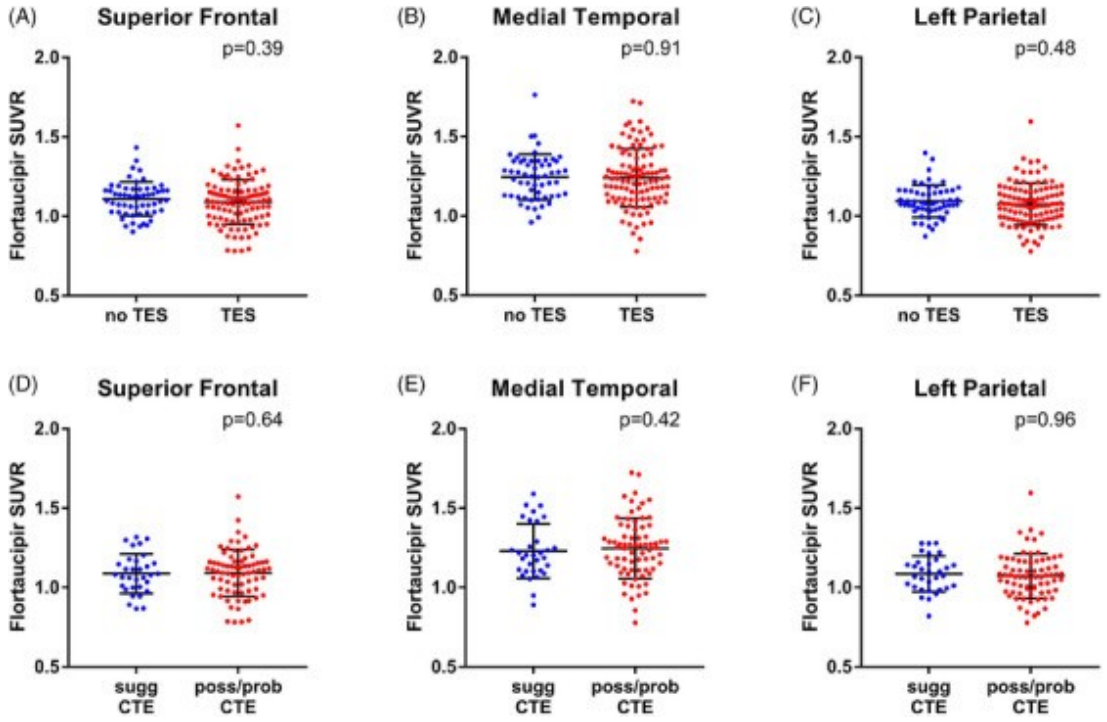
Play video

A group difference is not an individual diagnostic threshold

Two groups can have different average imaging values while their individual values overlap substantially. A result about the average therefore does not establish a boundary that accurately classifies every person. A diagnostic threshold must be evaluated using individuals whose relevant condition is known through an appropriate independent standard.

Sensitivity describes how often a procedure detects the condition among people who have it. Specificity describes how often it correctly excludes the condition among people who do not. Changing a threshold can improve one while worsening the other, especially when the measured distributions overlap in the population being assessed.

CTE presents an additional challenge because definitive confirmation currently requires brain tissue after death. A living-participant research classification is not identical to that tissue standard. Studies can test associations with symptoms or exposure while separately evaluating whether the same measure eventually predicts neuropathologically confirmed disease.



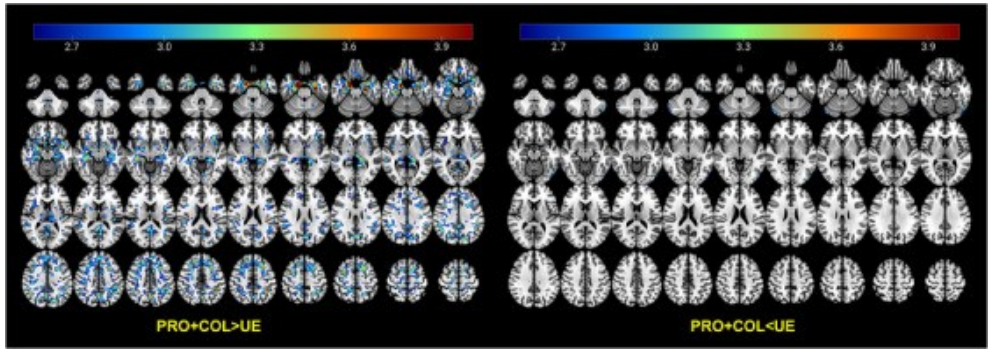
Su et al. (2024), Fig. 4.

Tau PET did not separate the clinical syndrome groups

The DIAGNOSE CTE project compared flortaucipir PET uptake in former professional players, former college players, and unexposed men. Former players had higher uptake in prespecified regions on group comparisons. An exposure association was observed in a superior-frontal region within the older player subgroup rather than uniformly throughout the sample.

The PET measure did not distinguish the traumatic encephalopathy syndrome groups examined in the study. This separates evidence of an average uptake difference from evidence of clinical classification. It also leaves the molecular interpretation dependent on validation of how flortaucipir binds the pathology present in CTE and other conditions.

The separate clinical recording demonstrates visual recognition, including identifying familiar faces and colors. It makes an observable cognitive ability distinct from tracer uptake. Neither the demonstration nor the PET group difference establishes the individual's CTE status; each measures a different feature that a broader clinical investigation may examine.



Su et al. (2024), Fig. 2.



Clinical demonstration: visual recognition. Larsen and Stensaas (2020).

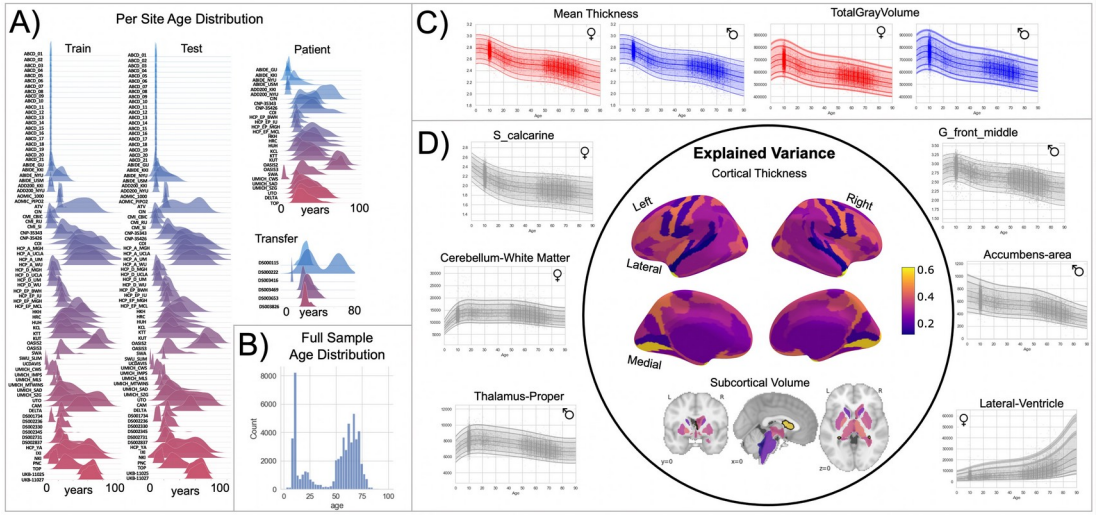
Play video

An imaging reference describes the comparison population

An imaging value acquires meaning through comparison with an appropriate reference. Age, acquisition site, and processing method can influence cortical thickness and regional volume. A value that differs from the average for young adults may be typical for an older person, so the chosen reference must match the clinical question.

Normative modeling estimates the range and trajectory of measurements within a reference population. It can describe how far an individual's value departs from the expected distribution under the model. This is different from treating every deviation as disease or assigning the same psychological consequence to every anatomical difference.

A brain-age estimate summarizes how an image resembles patterns associated with chronological age in training data. It does not count years of injury or prove that the person's brain has a specific disease. Its clinical usefulness depends on independent prediction of outcomes rather than the intuitive appeal of an older-looking numerical label.



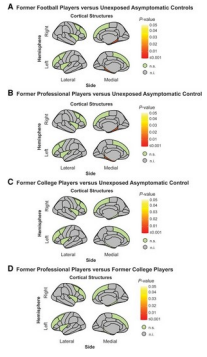
Rutherford et al. (2022), Fig. 1.

Regional MRI did not identify the clinical syndrome

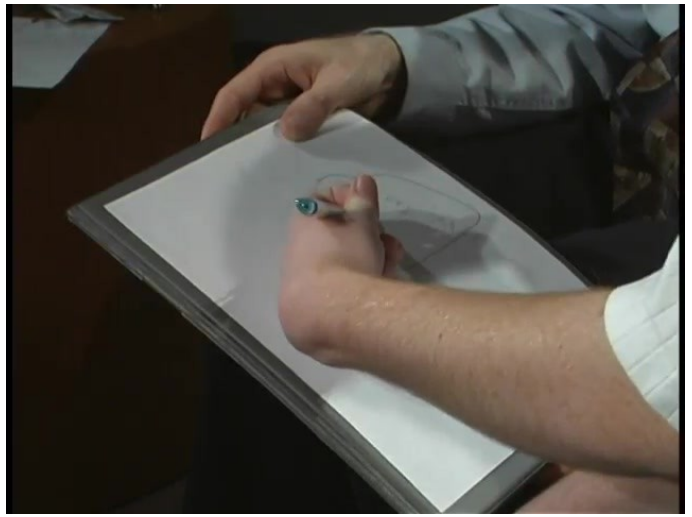
A structural MRI study measured regions selected because they are affected in CTE pathology reports. Former football players showed reductions in selected volumes or thicknesses compared with unexposed asymptomatic controls. Regions included the hippocampus, amygdala, entorhinal area, and parts of temporal and frontal cortex under the reported comparisons.

The regional measures did not show the same associations with the traumatic encephalopathy syndrome diagnosis, its core clinical features, or provisional CTE-certainty categories. Many exposure associations were also absent. The study therefore measured anatomical group differences without establishing that those measurements identified the clinical syndrome or a confirmed CTE lesion.

The separate recording shows a person having difficulty constructing a clock. The response describes a functional problem requiring further examination; it is not a demonstration of the MRI findings in a study participant. Structural images and observed task errors must be compared directly rather than assuming that either automatically determines the other.



Arciniega et al. (2024), Fig. 1.



Clinical demonstration: clock-drawing difficulty. Larsen and Stensaas (2020).

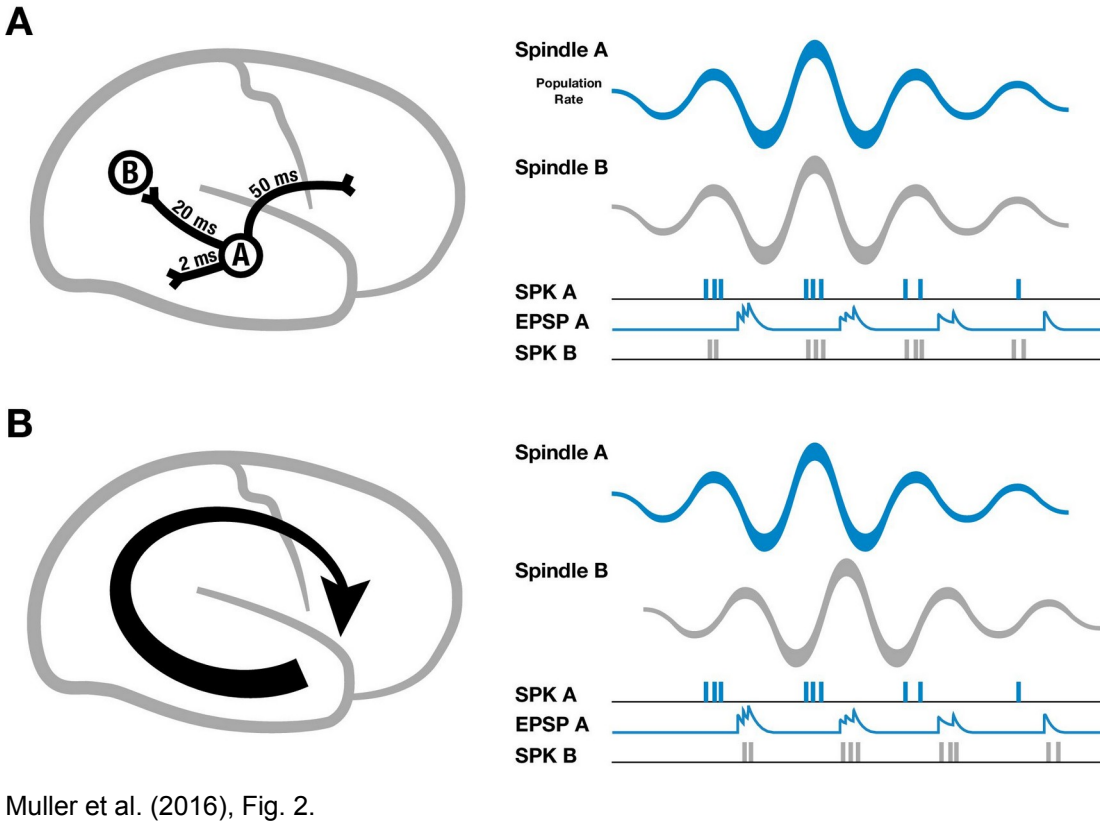
Play video

Sleep changes the conditions under which memory operates

Sleep is an active physiological state with patterns of coordinated brain activity. A sleep spindle is a brief burst of oscillation associated with interactions between thalamus and cortex. Spindles occur during non-rapid-eye-movement sleep and have been investigated in relation to the stabilization of newly learned information.

Consolidation describes processes through which a memory becomes more stable after learning. It is different from paying attention when information is first presented or retrieving it during a later examination. A person can therefore experience difficulty remembering for reasons involving several stages, including the conditions under which learning and sleep occur.

A clinical history asks about sleep quality, daytime sleepiness, and possible sleep disorders alongside cognitive complaints. Poor sleep can affect alertness during testing and everyday functioning. These observations concern potentially treatable contributors to difficulty and do not require assigning every complaint to an irreversible neurodegenerative process.



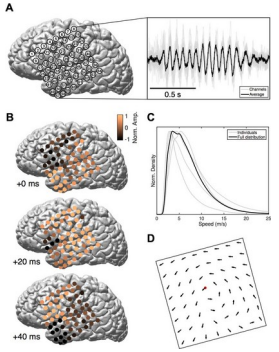
Muller et al. (2016), Fig. 2.

Human sleep spindles formed rotating cortical patterns

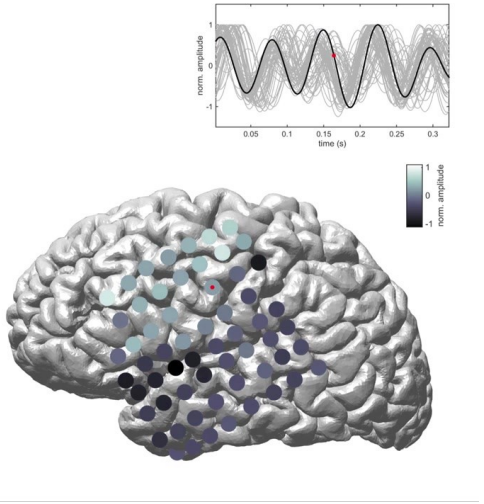
Researchers recorded electrical activity from electrodes on the human cortical surface during sleep. Rather than appearing everywhere simultaneously, spindle activity formed rotating wave-like patterns across the recorded regions. The timing of the activity varied systematically across locations within the measurement array during individual spindle events.

The original recording displays several spindle cycles using signal values positioned on a cortical surface. The authors also found highly similar patterns that recurred at separate times during the night. Repeated coordination at the measured time scale offers a proposed mechanism through which distributed cortical activity could participate in stabilizing memories.

The recording is an electrophysiological visualization, not an MRI scan or footage from a football study. Its colors encode the recorded signal under the authors' display procedure. It demonstrates that sleep-related coordination has a measurable temporal organization that is lost if brain function is described only through a static anatomical image.



Muller et al. (2016), Fig. 1.



Human cortical sleep-spindle activity. Muller et al. (2016).

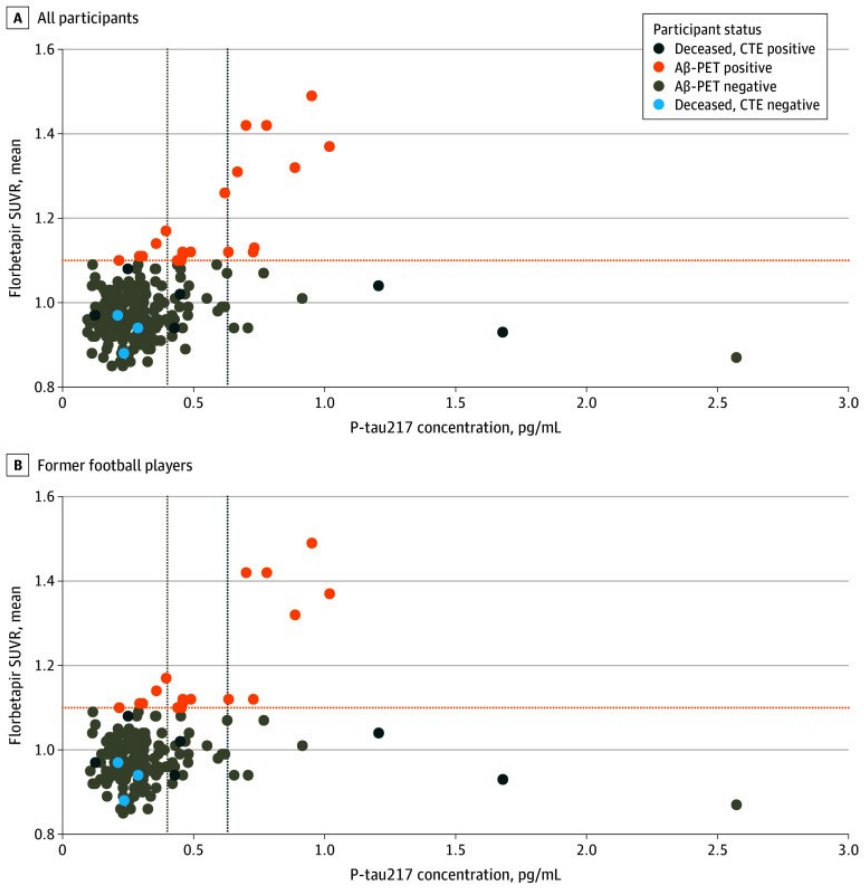
Play video

A blood tau marker reflected Alzheimer-related biology

A 2026 study followed blood phosphorylated tau 217 in former football players and unexposed participants. The investigators also examined amyloid PET and available pathology information. Former players had higher baseline concentrations on group comparisons, but amyloid-positive participants accounted for an important part of the difference.

The blood marker closely tracked amyloid imaging and Alzheimer disease status rather than estimates of repetitive-impact exposure. Concentrations increased over time without showing the expected exposure associations for a specific CTE marker. The same protein measurement therefore required a different disease interpretation from the one its study population might initially suggest.

Alzheimer disease can contribute to cognitive difficulty in a former player, just as it can in other adults. Identifying that contribution can change the clinical assessment and the treatments considered. A football history does not remove the need to investigate other diseases, and an Alzheimer-associated biomarker should not be renamed a CTE test.



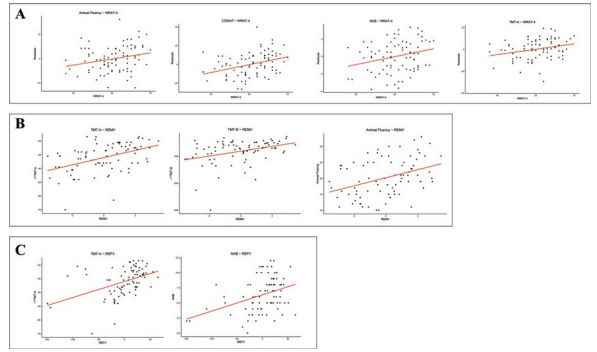
Miner et al. (2026), Fig. 2.

Reserve estimates depend on their measurement

The reserve study distinguished conventional indicators from performance remaining after measured demographic and brain variables were considered. Reading ability was associated with several cognitive domains, while education and occupational measures did not behave identically. Elite athletic careers can make some conventional occupation-based indicators difficult to interpret as comparable measures of cognitive opportunity.

A residual performance measure summarizes how well a person performs relative to the variables included in its model. It can change when the model includes different information. The study found associations between these reserve estimates and other cognitive abilities, describing variation in functioning rather than proving that reserve prevents a particular tau lesion.

The clinical recording demonstrates recognizing objects through touch with vision removed. The person must receive adequate sensory information, explore the object, and retrieve its identity. This separate examination illustrates why measured performance depends on several prerequisites; it is not one of the reserve computations or a recording of a player in the study.



Foley ÉM et al. (2022), Fig. 1.



Clinical demonstration: recognizing objects by touch. Larsen and Stensaas (2020).

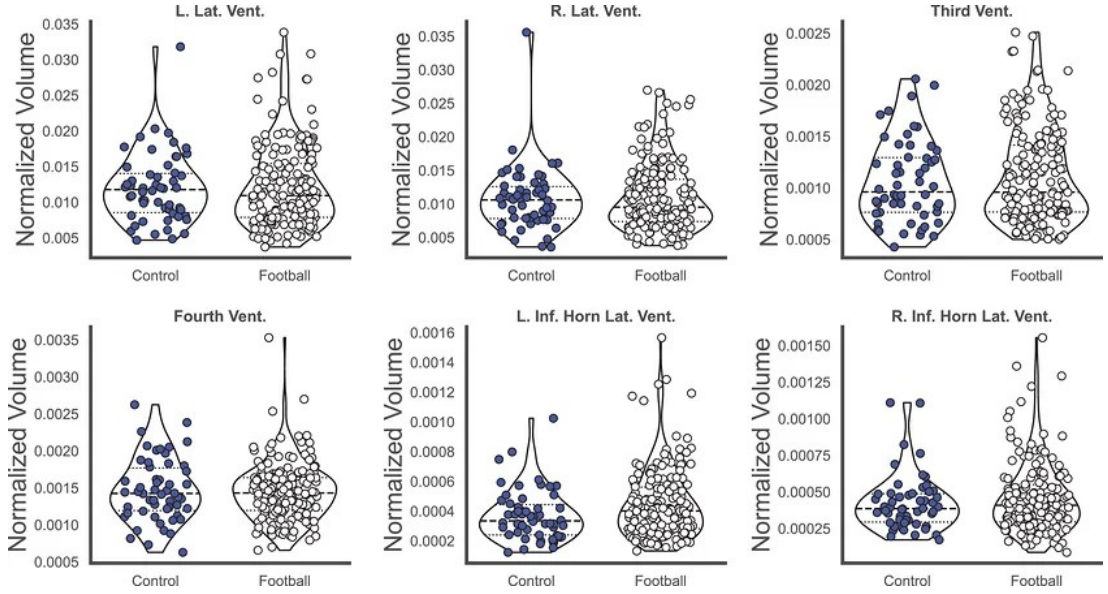
Play video

Ventricular differences tracked estimated exposure burden

A 2026 MRI analysis manually measured several ventricular regions in former football players and unexposed controls. Former players had larger left inferior-lateral ventricular volume, with the largest reported effects in professional players. The investigators examined regional measurements rather than assuming that all fluid spaces enlarged in the same way.

Greater estimated cumulative linear and rotational acceleration accompanied enlargement in lateral-ventricular and inferior-horn regions. These estimates summarized exposure history rather than a direct lifetime recording of every player's impacts. The anatomical findings concerned cumulative biomechanical loading and did not independently identify which underlying tissue processes produced the enlargement.

An enlarged space near memory-related structures can motivate a clinical question but cannot replace memory testing. The study's ventricular measurements and the person's daily functioning remain different outcomes. Follow-up can examine whether anatomical change precedes, accompanies, or fails to predict later decline under a specified measurement procedure.



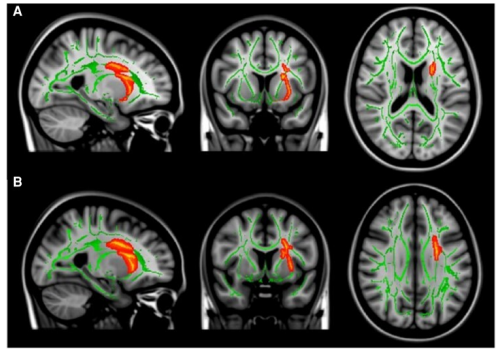
Szekely et al. (2026), Fig. 1.

White-matter diffusion showed different patterns

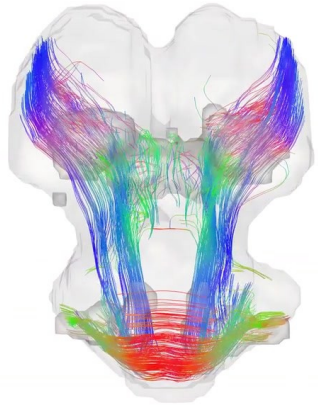
A DIAGNOSE CTE diffusion study compared former football players with unexposed asymptomatic controls. Players showed higher fractional anisotropy in a limited part of the white-matter skeleton on group comparisons. Other diffusion measures did not show the same group differences, and the measurements did not distinguish the clinical syndrome groups.

Within players, older age and earlier first exposure accompanied lower values on selected measures, while greater cumulative impact burden accompanied lower tissue-corrected anisotropy. These patterns show why anisotropy cannot be interpreted as a simple scale on which every higher value means healthier tissue or every lower value proves one specific injury process.

The separate original reconstruction shows living-participant auditory-pathway streamlines from another human MRI study. It demonstrates the geometry estimated by tractography, not the football cohort's result. A streamline image and a diffusion-value comparison are related products of imaging, but they summarize different features of the underlying measurements.



Arciniega et al. (2026), Fig. 1.



Human auditory-pathway tractography. Sitek et al. (2019).

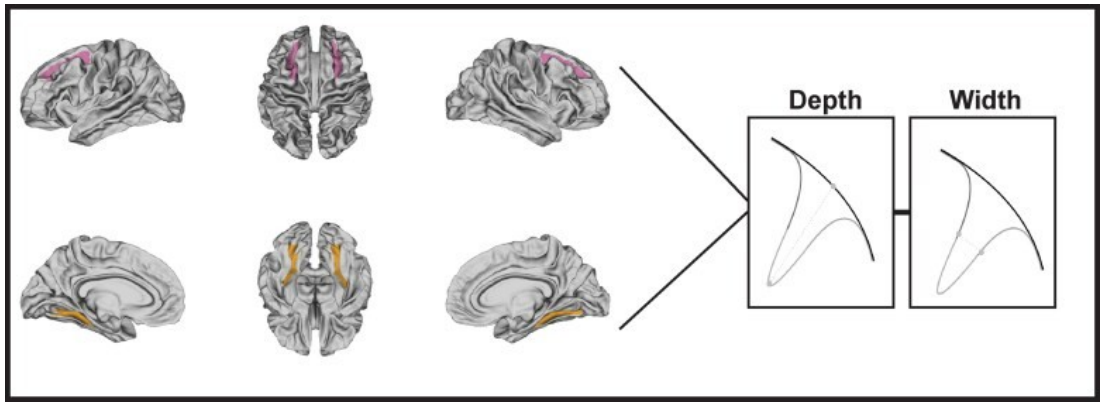
Play video

Sulcal geometry varied with exposure

Researchers measured the depth and width of selected cortical sulci in former football players. A sulcus is a groove between cortical folds. The superior-frontal and occipitotemporal regions were selected to examine whether their geometry differed between exposure groups and whether it accompanied clinical or imaging measures relevant to CTE research.

Former players had shallower depth in a left superior-frontal region than controls. Selected exposure measures accompanied reduced depth or greater widening in particular regions. However, sulcal geometry did not show the same associations with the clinical syndrome classification, neuropsychological performance, or flortaucipir uptake in the reported analyses.

The anatomical rationale originated partly from the location of CTE lesions at sulcal depths, but a rationale is not a demonstrated diagnostic correspondence. The study measured surface geometry in living individuals. Confirming that a particular geometric pattern identifies the characteristic tissue lesion would require evidence connecting the measurements to neuropathology.



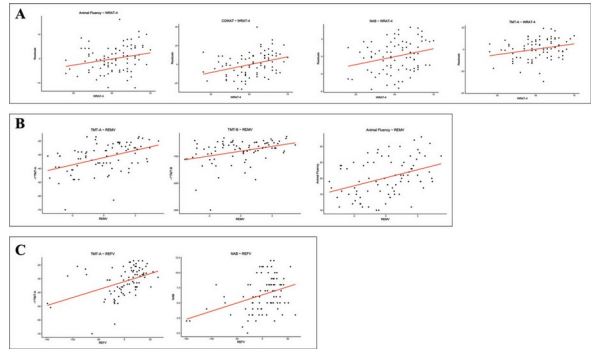
Jung et al. (2025), Fig. 1.

Verbal fluency reveals a pattern of responses

In the reserve study, reading ability accompanied both semantic and phonemic fluency. Semantic fluency requires generating items from a category, whereas phonemic fluency uses an initial-letter rule. These tasks share retrieval and executive demands but organize the search for words differently, allowing a more detailed account of performance.

The separate abnormal examination recording shows a person beginning category-word generation and then stopping after producing only a few items. The examiner can record initiation, sustained output, repetitions, and rule violations. A reduced total can therefore be described through the actual pattern of responses rather than only through the label poor fluency.

Language knowledge, retrieval efficiency, maintained attention, and the ability to organize a search can all contribute to the observed response. Comparing fluency with other tasks helps identify which operations require further assessment. The teaching recording is not evidence that the displayed person has CTE or belongs to the former-player reserve sample.



Foley ÉM et al. (2022), Fig. 1.



Clinical demonstration: limited verbal fluency. Larsen and Stensaas (2020).

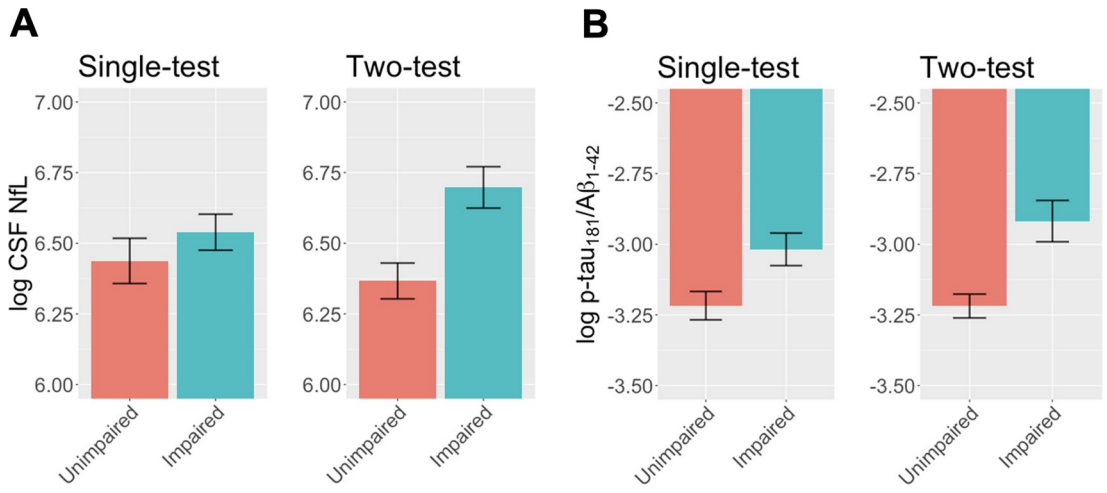
Play video

Repeated testing introduces practice effects

A cognitive score is an estimate obtained under particular conditions. Repeating a task can improve performance because the person recognizes its structure, remembers material, or adopts a better strategy. A later improvement can therefore include a practice effect rather than a complete change in the underlying cognitive ability.

A decline also needs interpretation relative to expected variability and the interval between assessments. Different versions of a test can reduce memory for specific items while preserving the ability being examined. Consistent administration makes comparison easier by limiting changes caused by instructions, equipment, or testing conditions.

Clinical follow-up combines these measurements with the person's history and daily functioning. A stable score, a changing complaint, and a new need for assistance may describe different aspects of the same period. The assessment should document each observation rather than forcing every source of information into a single presumed trajectory.



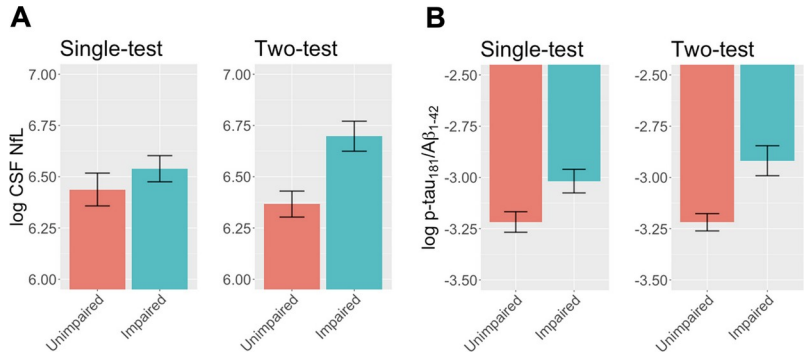
Ly MT et al. (2025), Fig. 1.

Two-test criteria selected a different clinical group

The cognitive-classification study required either one low test result or two lower results within a domain. The two-test approach identified fewer former players as impaired. Those classified by that approach also had greater exposure and selected fluid-marker differences that did not appear with the single-test classification in the reported analyses.

Requiring convergent evidence within a domain reduces the influence of an isolated unusual score, although it also changes who is included. The study examined this choice through associations with independently measured biomarkers. It did not establish that every person meeting either rule had CTE, because a clinical impairment category and tissue diagnosis are different outcomes.

The separate examination recording demonstrates right-left orientation, finger naming, and calculation. These tasks show how a battery can sample distinct operations rather than repeat one question in several forms. The clip is not the study's testing session or classification procedure; it makes the action behind a neuropsychological test item observable.



Ly MT et al. (2025), Fig. 1.



Clinical demonstration: calculation and orientation. Larsen and Stensaas (2020).

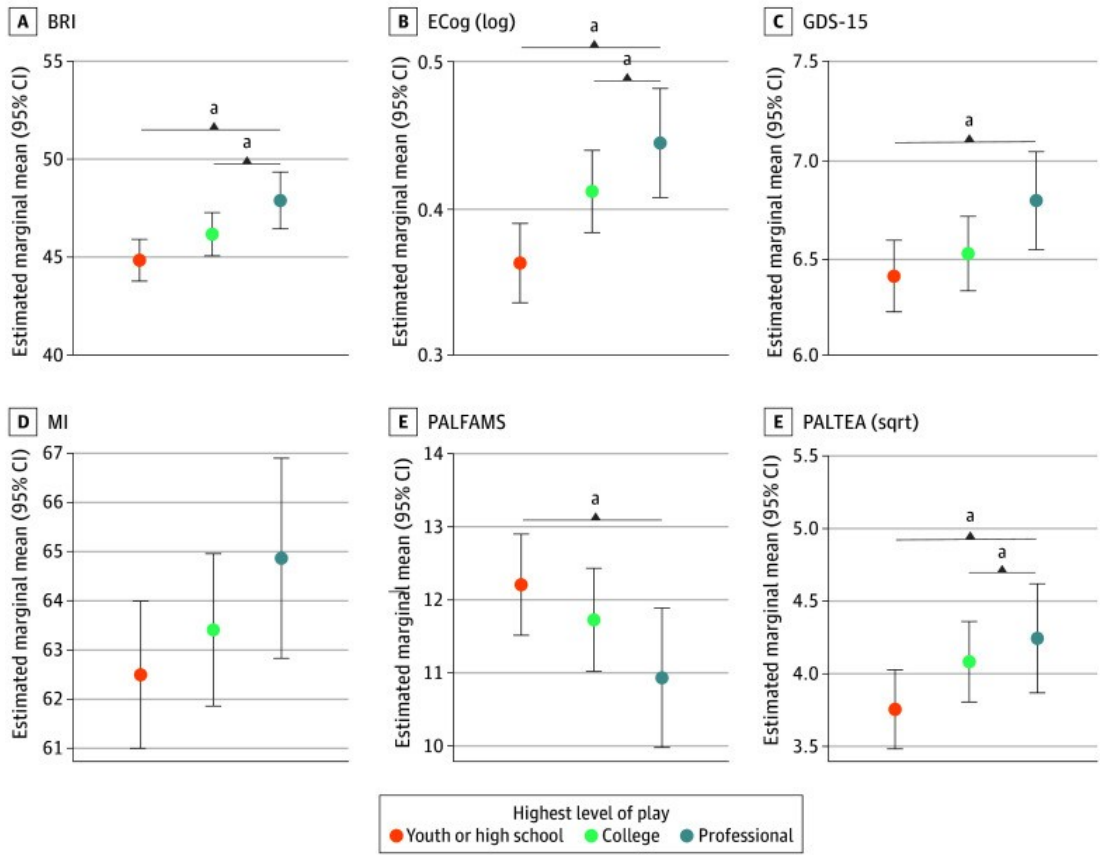
Play video

A complaint and a test answer describe different demands

A subjective cognitive concern describes what a person notices about their own functioning. A performance-based measure records what the person does during a specified task. Daily life may involve distraction, fatigue, competing goals, and emotional demands that differ from a quiet assessment with direct instructions and an examiner present.

The reverse difference can also occur. A person may perform below expectation on a task without noticing the same difficulty in ordinary life, particularly when familiar routines or other people compensate for the problem. Comparing the two observations asks what each measurement captures rather than declaring one automatically irrelevant.

An assessment establishes whether the complaint concerns learning, remembering, attention, speed, or another operation. It asks for examples, checks whether the pattern has changed, and examines consequences. These details allow treatment and follow-up to address the actual difficulty even when a definitive explanation of its underlying pathology is not yet available.



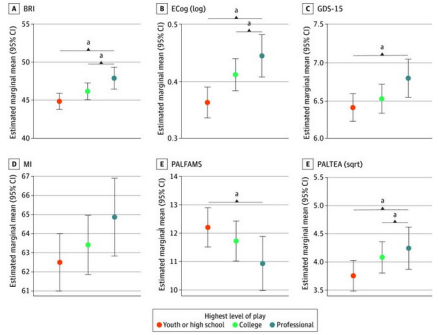
Aaronson et al. (2026), Fig. 2.

Player results included reports and performance

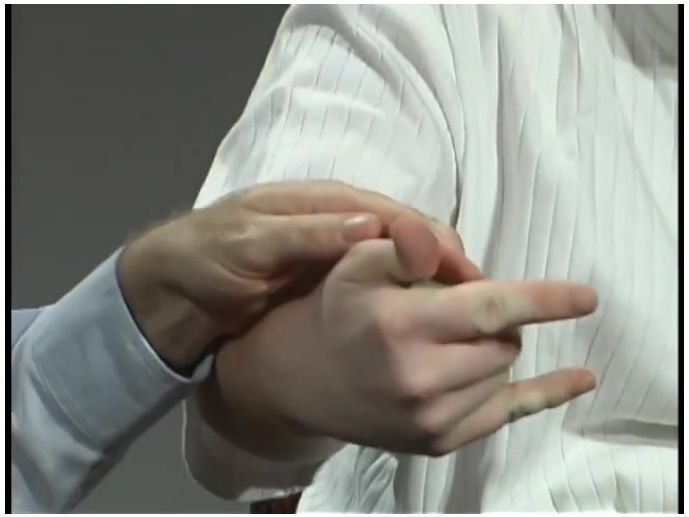
The large former-player study measured computerized performance alongside reported cognitive concerns and neuropsychiatric symptoms. Higher playing level and longer participation accompanied poorer outcomes within the study. A matched-control comparison also found differences in measured performance, subjective concerns, and depressive symptoms, treating them as related but separately recorded outcomes.

The observations were collected in a cross-sectional design, meaning that the comparison assessed people at the studied time rather than directly observing their entire earlier cognitive trajectory. Its results add evidence about functioning among former players while leaving the cause of any one participant's difficulties to further clinical investigation.

The separate clinical recording shows difficulty identifying objects through touch, with performance differing across hands and possible motor interference noted by the source. It demonstrates why an unexpected task response must be examined in context. A score can reflect sensory or motor prerequisites as well as the higher-level ability the examiner intended to assess.



Aaronson et al. (2026), Fig. 2.



Clinical demonstration: object-recognition difficulty. Larsen and Stensaas (2020).

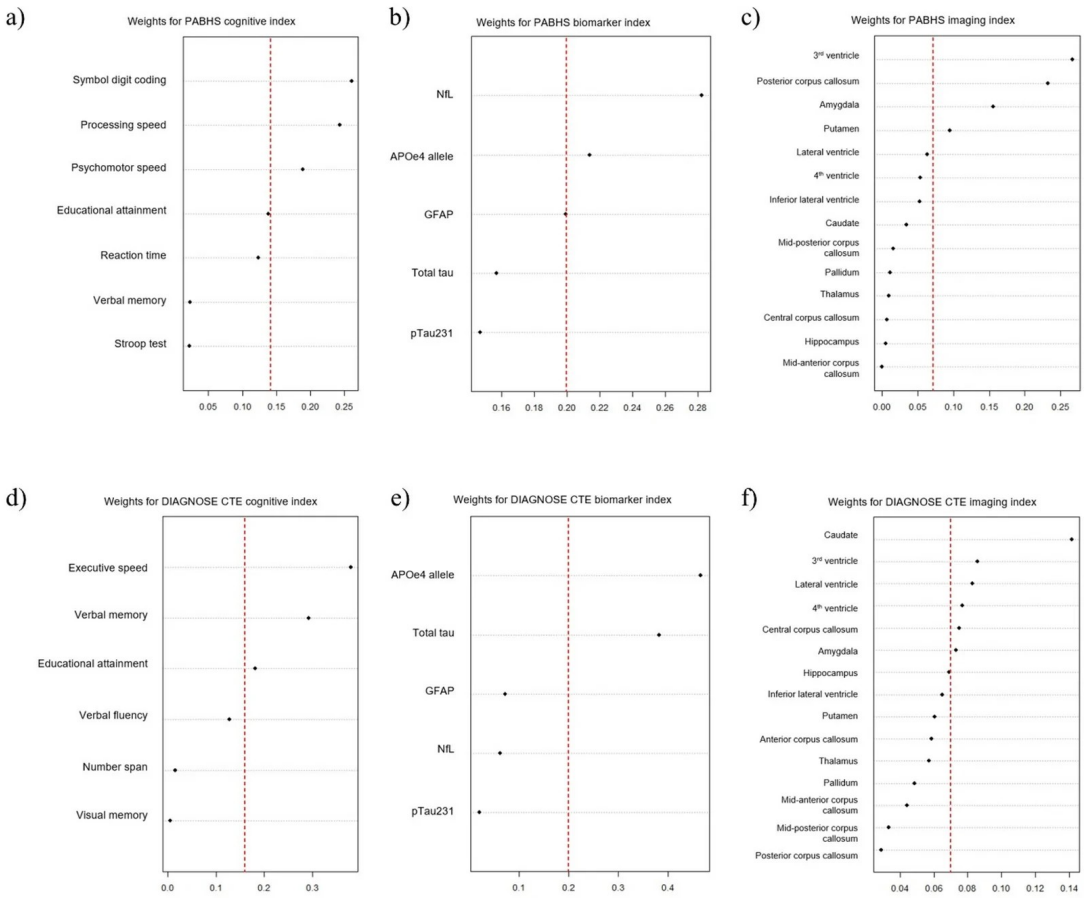
Play video

Combined measures require an independent outcome

A multimodal analysis combines information from different measurement types, such as cognitive tests, blood proteins, and MRI. The aim is to determine whether the combination describes an outcome better than one type alone. Each input must be defined clearly, because the same clinical information can otherwise enter both the predictor and the outcome.

If cognitive impairment helps define a syndrome, a model containing cognitive scores may partly recover that definition. An analysis must therefore distinguish classifying the clinical syndrome from predicting an independent tissue diagnosis or a later functional outcome. These are different targets even when the model produces a similar-looking probability for each.

Validation tests whether the model works beyond the observations used to construct it. A model may behave differently in fighters, football players, and a general clinic because exposure and other illnesses differ. The comparison population and the independently established outcome determine what a successful prediction would mean in practice.



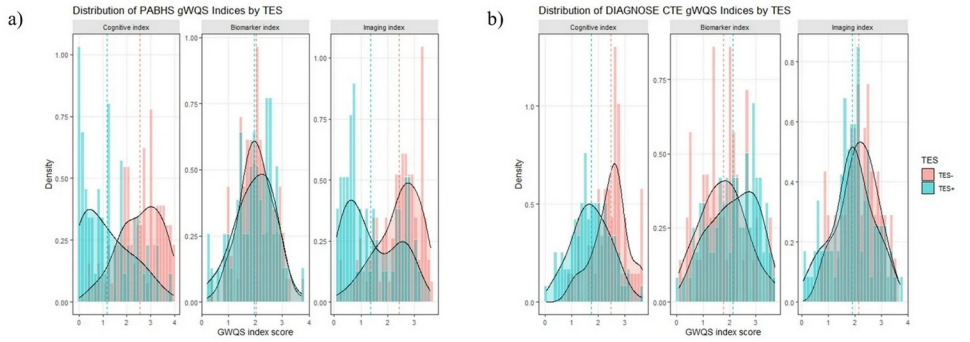
Conway Kleven et al. (2026), Fig. 1.

Several measurement domains contributed to TES models

A 2026 study examined former football players and professional fighters in separate cohorts. The investigators constructed cognitive, blood-biomarker, and structural-imaging indices and tested their combined association with traumatic encephalopathy syndrome. The analysis compared a clinical research classification with information drawn from several biological and behavioral measurement domains.

Combined models performed better than individual-domain models in the reported analyses. Imaging contributed more prominently in the fighter cohort, while blood-marker associations were stronger in the football cohort. The result demonstrates that contributions can vary across populations rather than establishing a universal weighting that applies to every collision-sport athlete.

The separate recording demonstrates practical judgment and abstract reasoning through verbal questions. It makes an executive demand observable but is not a participant record from either cohort. The study predicted a clinical syndrome classification; it did not replace the need for pathology-based validation before a combined score could establish CTE during life.



Conway Kleven et al. (2026), Fig. 3.



Clinical demonstration: judgment and reasoning. Larsen and Stensaas (2020).

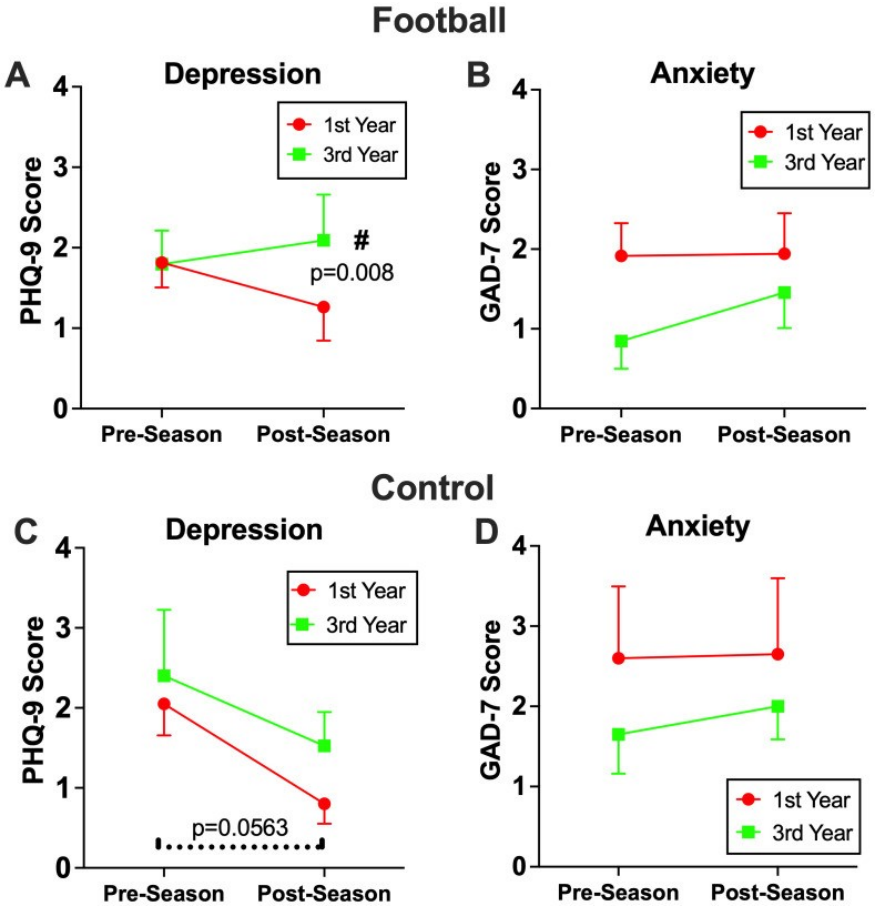
Play video

Depression can affect cognition without defining its cause

Depression can involve reduced motivation, disrupted sleep, fatigue, slowed activity, and difficulty concentrating. These changes can interfere with learning and completing tasks. A complaint about poor memory may therefore include problems with attention or effortful engagement, even when the person describes the experience primarily as forgetting.

Mood and cognitive disorders can also coexist. Identifying depressive symptoms does not prove that every difficulty is explained by mood, just as identifying a football history does not prove that every symptom is caused by CTE. The clinical task is to determine which conditions are present and what each contributes to the person's functioning.

Assessment includes symptom duration, changes in interests and routine, and effects on daily activities. Treatment can address identified psychiatric symptoms while cognitive evaluation continues. A person does not need a confirmed CTE diagnosis before clinically significant depression, anxiety, pain, or sleep difficulty deserves direct attention and appropriate care.



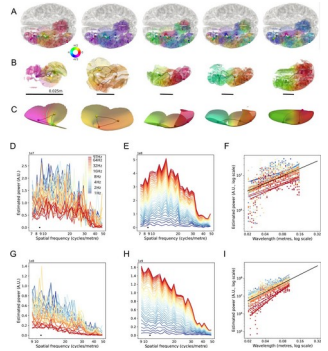
Buddenbaum et al. (2025), Fig. 4.

Electrical timing adds information beyond activation

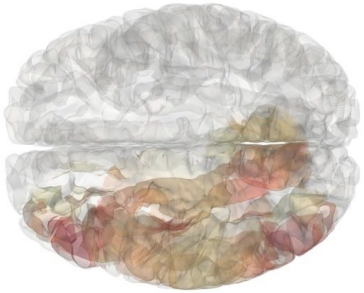
Researchers used intracranial electrical recordings to examine how oscillation phase varies across widely separated human cortical sites. Phase describes where a signal is within an ongoing cycle. The study analyzed patterns over large spatial scales rather than considering only the strength of activity at one electrode or anatomical location.

The original visualization shows a low-rank reconstruction of phase dynamics in the delta-frequency range. Its colors encode the authors' phase display, with interpolation between recording contacts. The investigators found especially strong contributions from long spatial wavelengths across several temporal-frequency ranges, describing organized patterns that extend over large cortical distances.

This is a mathematical visualization of human electrical recordings, not a colored MRI lesion map or a recording from a football cohort. It shows why network timing is a distinct measurement dimension. Two regions can have similar average activity while differing in the temporal organization of their ongoing signals.



Alexander et al. (2026), Fig. 5.



Human cortical phase dynamics. Alexander et al. (2026).

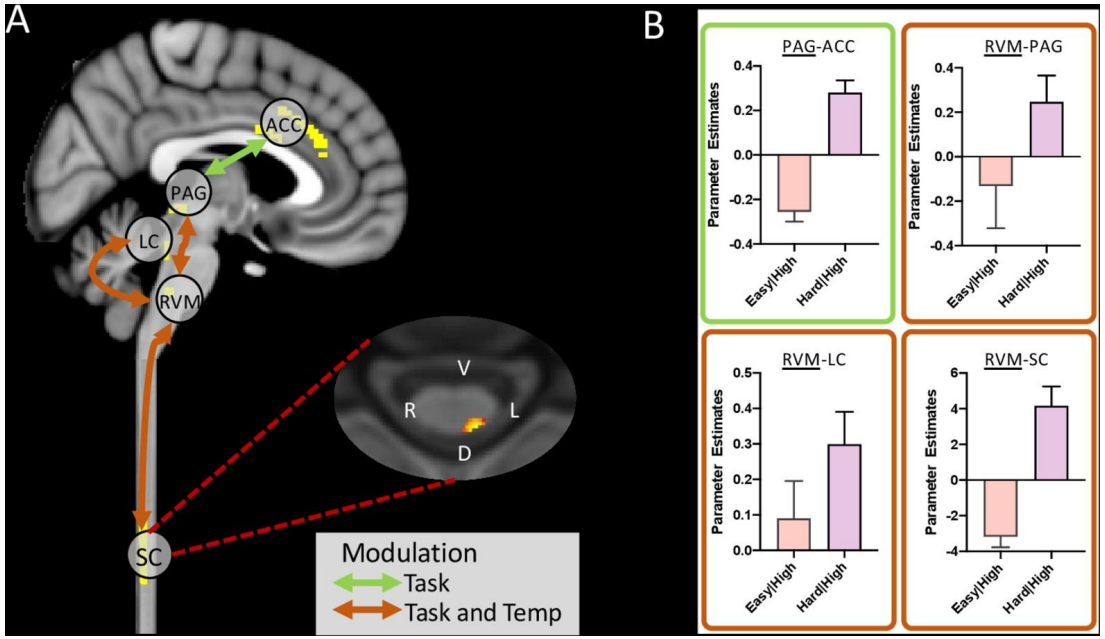
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Sleep, pain, and medicines belong in the same history

A person's cognitive performance can vary with current alertness, discomfort, and medication effects. Poor sleep can reduce sustained attention, while pain can repeatedly capture attention during a task. Some medicines can also produce sedation or other cognitive effects, so the timing and purpose of each medication need to be reviewed.

These contributors can coexist with structural brain changes or a neurological disease. An assessment records when symptoms began relative to changes in sleep, pain, or treatment. This timeline helps identify potentially reversible influences without assuming that finding one contributor rules out every additional condition requiring investigation.

The research criteria for traumatic encephalopathy syndrome require considering whether other conditions fully account for the clinical features. In ordinary care, identifying a treatable contributor also supplies an immediate target for management. Follow-up can examine whether functioning changes when that contributor is addressed under appropriate clinical supervision.



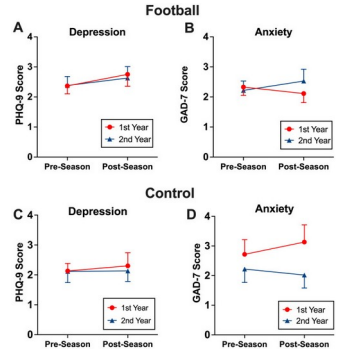
Oliva et al. (2022), Fig. 4.

Starting age did not predict symptoms in every sample

An online study of men who had played high-school football compared earlier and later ages of first exposure. Participants reported health history and current cognitive and psychological symptoms. Men who began earlier reported more lifetime concussions, but the exposure groups did not differ on the measured symptom ratings in that sample.

The analysis did not find associations between starting age and its current depression or post-concussion-like symptom measures. These outcomes differ from brain pathology or the MRI measures examined in other studies. A variable can therefore behave differently across selected populations and measurement procedures without the results answering exactly the same clinical question.

The Oregon and Washington State recording shows actual modern plays with players in different roles and contact situations. It provides sporting context rather than the study's self-reported histories or symptom data. The age at which someone began playing cannot be determined from the clip and cannot be used alone to assign a later diagnosis.



Buddenbaum et al. (2025), Fig. 3.



Oregon–Washington State football plays. FOX Sports (2018).

YouTube source

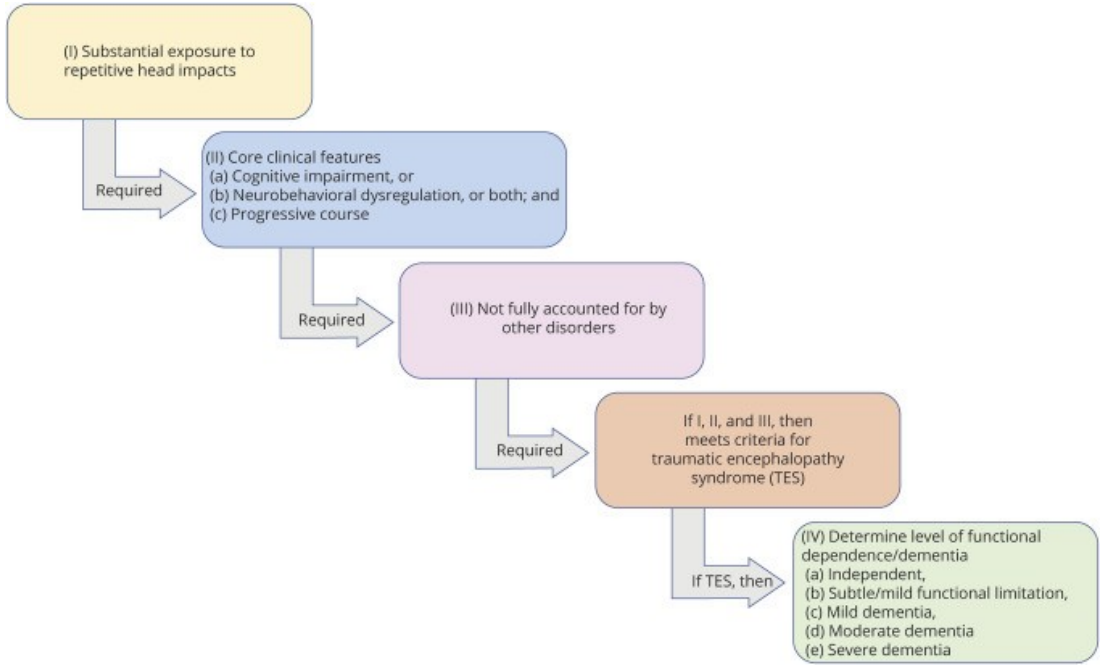
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Daily independence includes more than basic movement

Functional independence concerns whether a person can carry out the activities required in ordinary life. Basic activities include dressing, washing, and eating. Instrumental activities include managing finances, arranging transportation, keeping appointments, and taking medicines correctly. These categories place different demands on memory, planning, perception, and movement.

A person may move normally and converse fluently while needing substantial help with a complicated routine. Another person may need physical assistance while retaining the ability to plan and make decisions. Evaluating the actual task and the reason for assistance prevents physical dependence from automatically being interpreted as global cognitive decline.

Information from the person and from someone familiar with their routine can help identify changes. The history should record which tasks are affected, what assistance is required, and when the need began. This description supplies a functional outcome that can be followed even when cognitive test scores or imaging measurements remain relatively stable.



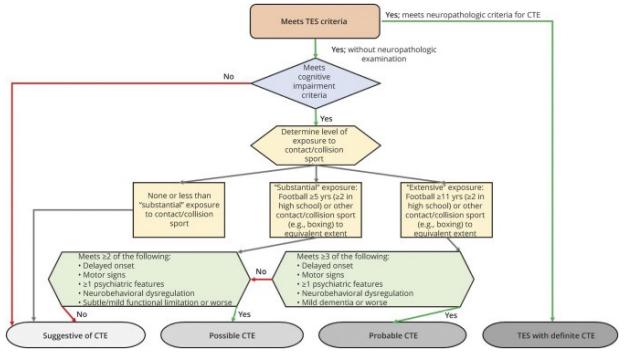
Katz DI et al. (2021), Fig. 1.

Research criteria graded dependence separately

The consensus process for traumatic encephalopathy syndrome distinguished the core clinical features from the severity of functional dependence. After determining whether the syndrome requirements are met, the framework grades dependence across several levels. The method preserves a distinction between having a cognitive or behavioral feature and needing extensive assistance in daily life.

The selected clinical recording demonstrates praxis, the production of familiar skilled actions in response to instructions. A person must understand the request and organize an appropriate movement using an imagined or represented object. Successful performance samples a specific ability rather than proving independence in every complex activity outside the examination.

The recording is a clinical teaching example, not a football participant or a direct demonstration of the consensus dependence categories. Asking about medication, driving, or finances adds a different kind of evidence. An assessment must determine how tested abilities operate together in the person's real environment and available support system.



Katz DI et al. (2021), Fig. 2.



Clinical demonstration: skilled actions. Larsen and Stensaas (2020).

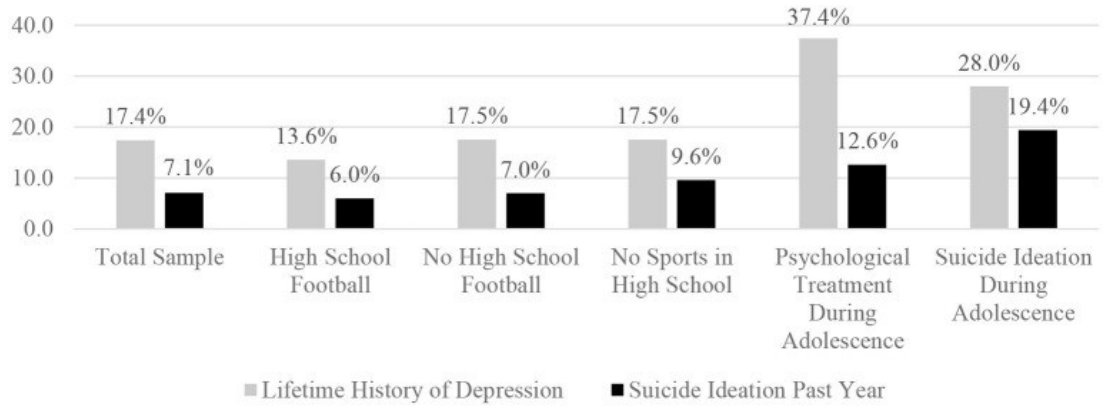
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Earlier mental-health history predicted later difficulties

In the national adolescent-to-adult cohort, mental-health difficulties during adolescence predicted later depression-related outcomes more clearly than high-school football participation did. The investigators could examine earlier history because participants had been followed over many years. This design supplied temporal information absent from a survey administered only after retirement.

A symptom reported during adolescence cannot automatically be treated as a new consequence of a later professional career. Conversely, a person with an earlier psychiatric history can still develop additional cognitive or neurological difficulties. The sequence and course of symptoms help distinguish continuing problems from new changes requiring separate evaluation.

The study did not show an increased adult mood-risk pattern for the high-school football group on its measured comparisons. Its result complements investigations of symptomatic former professionals by examining a different population and exposure context. Clinical interpretation needs the actual history rather than assuming one set of findings describes every level of football participation.



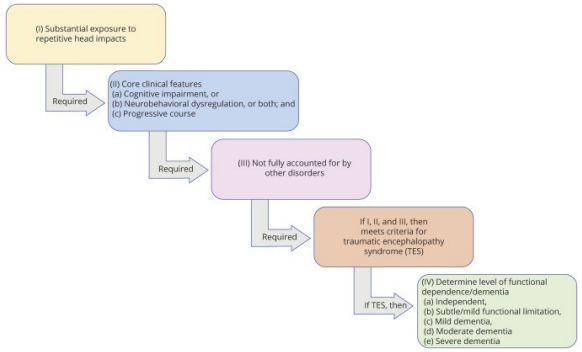
Iverson et al. (2021), Fig. 1.

A skilled-action error needs a specific description

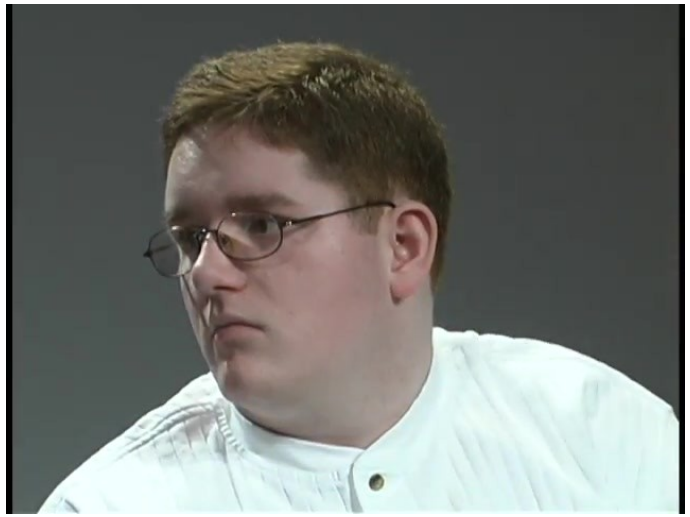
The consensus framework requires that clinical features be described and evaluated rather than inferred from a general complaint. Functional difficulty can arise through several operations, including understanding instructions, retrieving an action sequence, and organizing movements. Recording which step fails helps determine the further examination that is needed.

The clinical demonstration shows a person completing several requested actions but using the fingers as the blades of scissors when asked to demonstrate cutting. The source describes this as a possible body-part-as-object error. The behavior is more specific than saying the person cannot move or has lost all practical knowledge.

This separate teaching case is not evidence of CTE or a finding in a former player. It demonstrates why a clinical record preserves successful responses alongside errors. Comparing comprehension, voluntary movement, and action organization helps identify whether an apparent failure reflects language, motor ability, or the production of a learned skilled action.



Katz DI et al. (2021), Fig. 1.



Clinical demonstration: a skilled-action error. Larsen and Stensaas (2020).

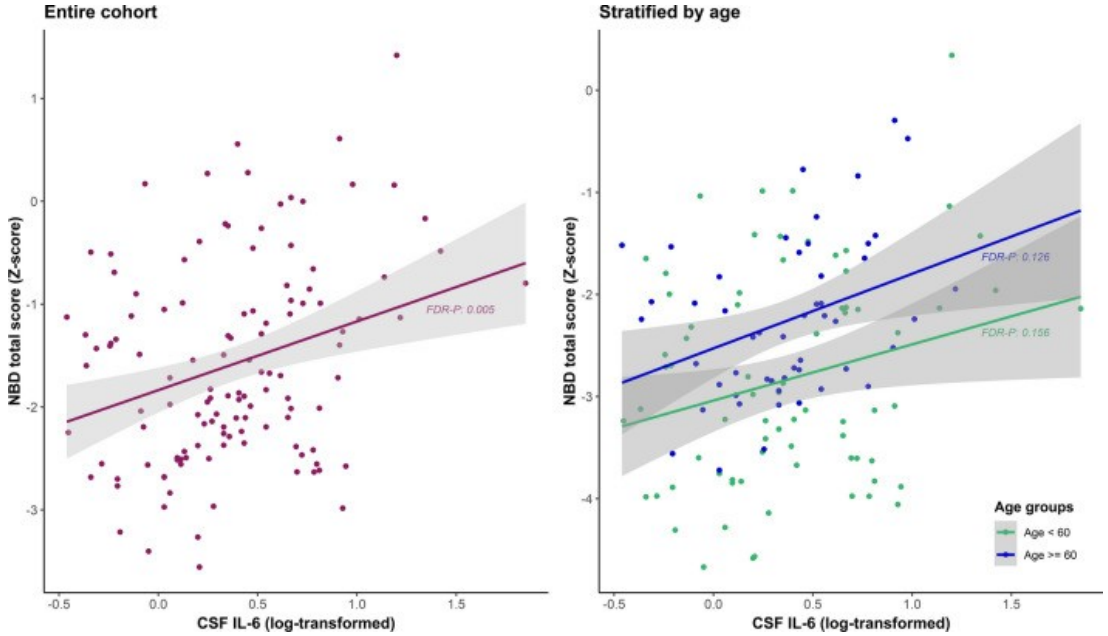
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A biological correlate is different from a treatment target

The inflammatory-biomarker study associated cerebrospinal-fluid interleukin-6 with selected dysregulation dimensions in former players. It did not experimentally reduce interleukin-6 and then demonstrate improvement in those symptoms. The measured association therefore identified a process for further investigation rather than an already established treatment for the clinical problem.

A treatment trial must specify an intervention and compare outcomes under adequate control conditions. Psychological outcomes can include the frequency of emotional episodes, the ability to maintain routines, and changes in relationships or work. Biological target engagement and meaningful improvement in functioning are related questions but require different measurements.

The same distinction applies when a player reports improvement after a treatment. A personal account can describe an important experience, while the reason for improvement remains a separate research question. Trials help distinguish the intervention's effect from symptom fluctuation, expectations, concurrent care, and other changes during the treatment period.



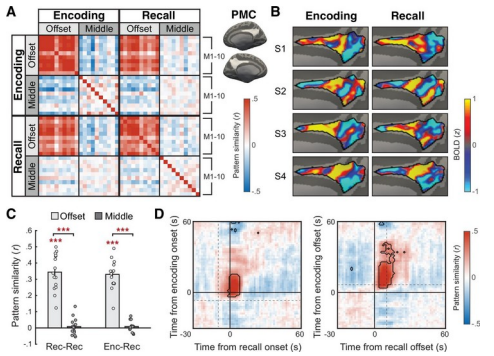
van Amerongen et al. (2024), Fig. 1.

Recall changed activity at mental-context boundaries

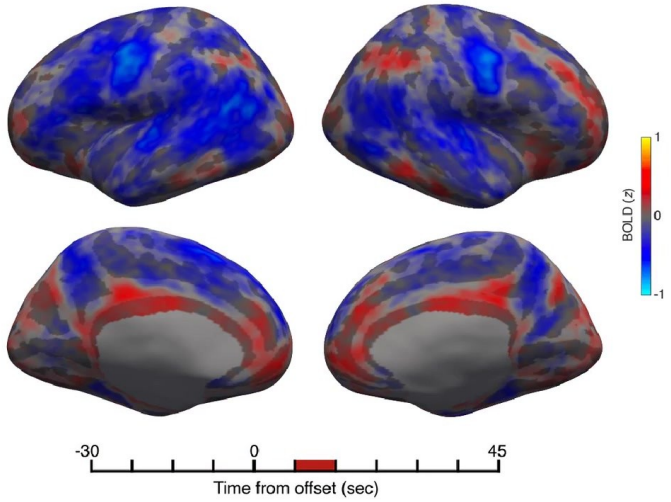
In a human fMRI study, participants watched several short movies and later recalled them aloud in an order they chose. Researchers examined moments when recall of one movie ended and another mental context began. Those transitions occurred without the external movie changes that had originally marked boundaries during viewing.

The investigators observed a recurring spatial activation pattern in the default-mode network, particularly posterior-medial cortex, after recall of a movie ended. The pattern was shared across different contents and between viewing and recall. It differed from patterns at boundaries between events occurring within a single movie.

The original visualization shows reconstructed activation changes around between-movie boundaries during recall. It is an analysis of human fMRI measurements rather than a recording of remembered images themselves. Researchers measured the organization of remembering through both what a person said and the timing of changes in the recorded network signals.



Lee et al. (2022), Fig. 3.



Human fMRI activity at recall boundaries. Lee et al. (2022).

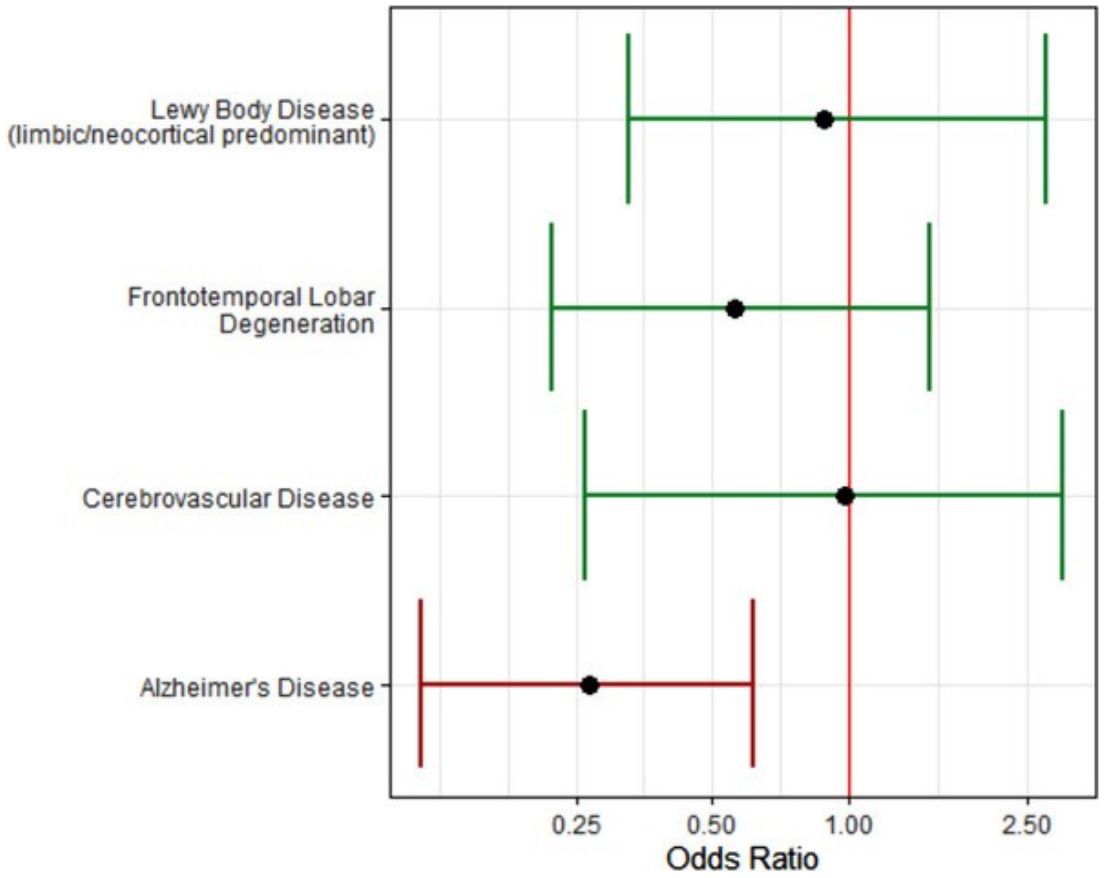
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Competing pathology changed clinical prediction

The brain-donor validation study found that the older syndrome criteria frequently classified people without CTE as having the clinical syndrome. Cognitive features were more informative about the tissue finding than the broad mood or behavioral categories examined. The presence of Alzheimer disease pathology further reduced the accuracy of the clinical classification.

The investigators compared clinical judgments and tissue assessments performed without access to the other result. This blinding reduced the chance that knowing a diagnosis would influence the interpretation of a history or slide. It allowed the researchers to evaluate how well the clinical information actually corresponded to independently assessed pathology.

For a living former player, investigating Alzheimer-related biology can therefore address a competing explanation for cognitive decline. The result does not mean that every memory problem is Alzheimer disease or that CTE and another pathology cannot coexist. It means that a clinical prediction must be evaluated against the conditions present in the person being assessed.



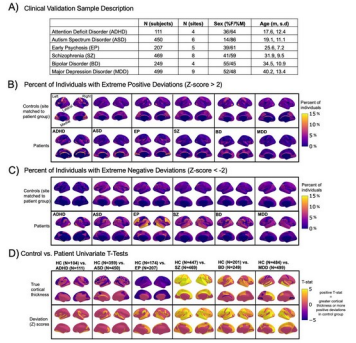
Mez et al. (2021), Fig. 3.

Normative brain models described individual variation

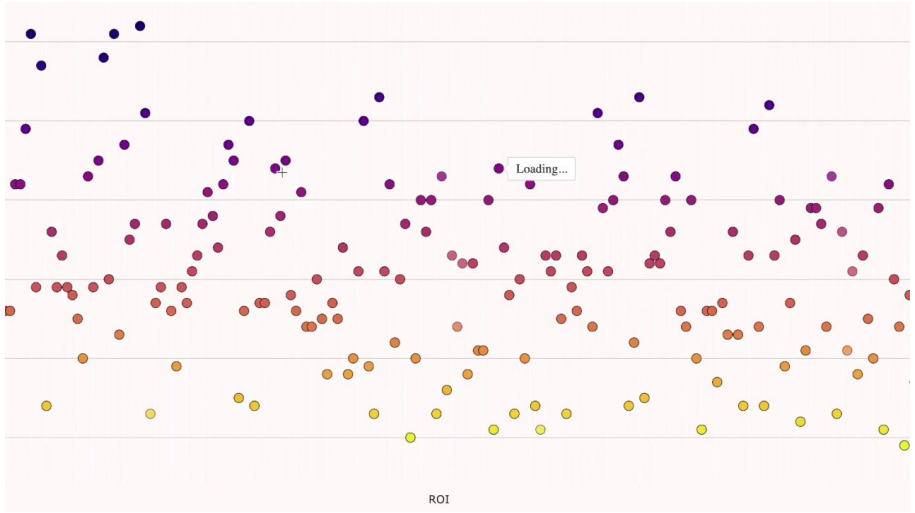
Researchers assembled structural MRI data across many sites and a broad age range to model cortical thickness and subcortical volume. Rather than defining one average healthy brain, the models described expected trajectories and variation. The investigators also evaluated a quality-checked subset and procedures for applying the models to data from new sources.

The original movie demonstrates an interface for examining the model's outputs. A reference trajectory allows an observed value to be compared with a distribution appropriate to the model's inputs. The paper applied the approach to psychiatric data to examine variation across diagnoses rather than assuming that every person within a diagnosis shares an identical anatomical pattern.

These models were not validated as individual CTE diagnostic tests. Their contribution is a method for describing variation and testing whether deviations predict an independently measured outcome. A player's image would still need appropriate acquisition, quality control, population comparison, and clinical interpretation before any deviation could guide a decision about that person.



Rutherford et al. (2022), Fig. 2.



Normative brain-imaging model interface. Rutherford et al. (2022).

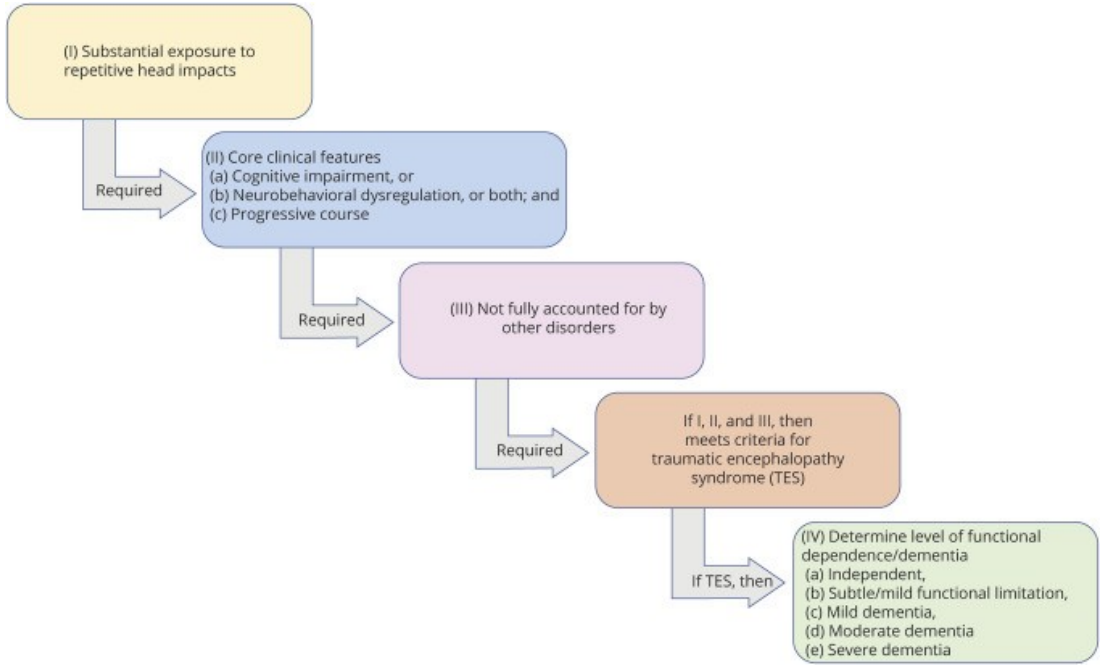
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Care can address functioning before pathology is known

A clinical plan begins with the difficulties the person is experiencing and the conditions that can be identified during life. Depression, anxiety, disturbed sleep, pain, movement problems, and cognitive impairment can each require attention. Their assessment and care need not wait for a definitive postmortem explanation of whether CTE is present.

Support can also change the demands of everyday tasks. Written reminders, consistent routines, reduced distraction, and appropriate assistance can help a person manage a specific difficulty. The choice depends on the observed problem and the person's goals rather than the assumption that every former player requires the same program.

Follow-up should examine meaningful outcomes, such as keeping appointments, managing medicines, sustaining relationships, or returning to a valued activity. It can also monitor whether symptoms progress, improve, or remain stable. These observations provide evidence about the person's course while research continues to evaluate CTE-specific diagnostics and disease-modifying treatments.



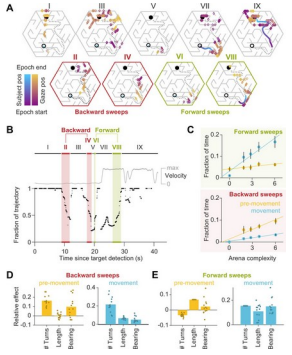
Katz DI et al. (2021), Fig. 1.

Eye movements revealed planning through a maze

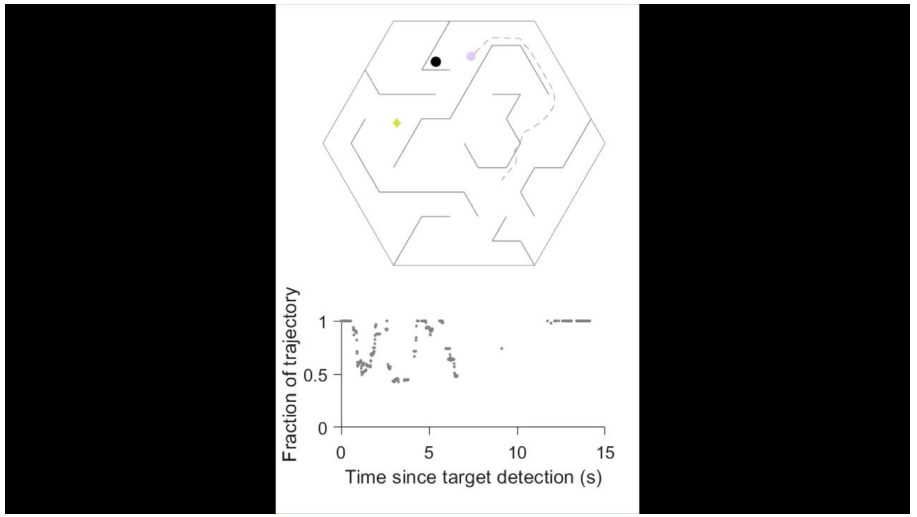
A human navigation study presented virtual mazes with briefly visible goals and recorded where participants looked while planning movement. As maze complexity changed, participants shifted attention between the goal and the transitions needed to reach it. The spatial and temporal pattern of gaze made parts of the planning process observable.

The original recording displays representative trials with sweeping eye movements. Researchers found sequences of looks that anticipated the future path rather than simply remaining fixed on the destination. These observations concerned visually guided planning under controlled conditions, with task structure providing a way to interpret the recorded eye movements.

Football also requires anticipating possible paths while monitoring a changing environment, but this experiment did not test players or diagnose injury. It provides a concrete cognitive-neuroscience method: specify the goal, control the information, and measure the actions used to plan. Clinical evaluation can then ask which component of performance has changed for a particular person.



Zhu et al. (2022), Fig. 4.



Eye movements during human route planning. Zhu et al. (2022).

Play video

Key takeaways

Describe the changed ability. Attention, episodic memory, executive control, mood, behavior, and independence require distinct observations and a timeline of change.

Compare complaints with performance. Reported difficulty, direct testing, and family observations describe complementary aspects of functioning; sensory and motor prerequisites can change task responses.

Identify what the scan measures. Structural MRI measures anatomy, diffusion MRI estimates water-motion properties, fMRI measures blood-oxygenation changes, and PET records tracer distribution.

Separate group results from diagnosis. Average differences between former players and controls do not establish an accurate individual CTE test. Clinical syndrome criteria and confirmed tissue pathology are different standards.

Examine competing and coexisting conditions. Alzheimer disease, Parkinson disease, depression, sleep disturbance, pain, and medication effects can contribute to symptoms in a former player.

Follow meaningful clinical outcomes. Care can address symptoms and daily functioning while pathology remains uncertain. CTE-specific diagnostic and treatment claims require independent validation and appropriate clinical evidence.